

★ WEEK 11 · PART 1 of 3

Continuous-Time Models: Brownian Motion and SDEs

When Time Flows Continuously

Brownian motion · the increment dW · SDEs · GBM · Ito's lemma

Every model so far ticked in discrete steps. Now we let time flow continuously — the language of modern finance. It begins with Brownian motion and the stochastic differential equations built on it.

1

PART 1 · BROWNIAN MOTION & SDEs

The Language of Continuous Time

Why continuous time, Brownian motion and its increment dW , stochastic differential equations, geometric Brownian motion for prices, and Ito's lemma — the stochastic chain rule.

The Thesis 60 Years Ahead of Its Time — Part 1

A student models the market with random motion

In 1900, Louis Bachelier's PhD thesis modeled stock-price fluctuations as what we now call Brownian motion — five years before Einstein's famous physics paper on the same phenomenon.

What he did

- Treated price changes as accumulating random increments.
- Derived the mathematics of Brownian motion for markets.
- Anticipated much of modern option pricing.

The reception

- It was far ahead of its time and largely overlooked.
- The tools to appreciate it did not yet exist.
- It sat mostly unread for half a century.

The Thesis 60 Years Ahead of Its Time — Part 2

Rediscovery and the continuous-time revolution

Rediscovered mid-century — notably by Paul Samuelson — Bachelier's idea, refined into geometric Brownian motion, underpinned the Black-Scholes-Merton revolution and modern continuous-time finance.

The revival

- Economists rediscovered the thesis in the 1950s.
- Brownian motion became the base for asset-price models.
- It powered the option-pricing breakthroughs of the 1970s.

The lesson for us

- Continuous time is the native language of modern finance.
- Its foundation is a single object: Brownian motion.
- From it we build SDEs — and a clean mean-reversion model.

From Discrete Steps to Continuous Flow

Where Weeks 1-10 leave us

1

Everything so far ticked discretely

ARIMA, GARCH, VAR, Kalman, HMMs — all evolve one step at a time, on a fixed grid of observations.

2

But markets don't wait for the bell

Prices move continuously; derivatives are priced in continuous time. We need models that flow rather than step.

3

Continuous time needs new tools

Brownian motion, stochastic differential equations, and Ito's calculus — the machinery of continuous-time finance, and the home of the OU mean-reversion model.

Part 1 in one line: build the continuous-time language — Brownian motion, SDEs, and Ito.

Why Move to Continuous Time?

Markets don't really wait for the closing bell

Discrete models pretend the world updates on a fixed clock. But trades happen at every instant, and the whole theory of derivatives lives in continuous time. Letting time flow gives cleaner, more powerful models.

What discrete time misses

- Prices move between the grid points, not just on them.
- Option pricing and hedging are continuous-time ideas.
- The math is often cleaner in the continuous limit.

What continuous time buys

- Elegant models with closed-form results.
- A natural home for calculus-based reasoning.
- The framework the rest of quantitative finance uses.

Brownian Motion (the Wiener Process)

The source of continuous-time randomness

$$W_0 = 0, \quad W_t - W_s \sim \mathcal{N}(0, t - s), \quad \text{independent increments, continuous paths}$$

Its properties

- It starts at zero and moves continuously.
- Increments over disjoint intervals are independent.
- An increment over length h is Gaussian with variance h .

Why it is fundamental

- It is the continuous-time limit of a random walk.
- It is the raw randomness every SDE is built from.
- Its paths are continuous but nowhere smooth.

The Increment dW

The atom of stochastic calculus

$$dW_t \sim \mathcal{N}(0, dt) \quad \Rightarrow \quad (dW_t)^2 = dt \quad (\text{the key heuristic})$$

What dW is

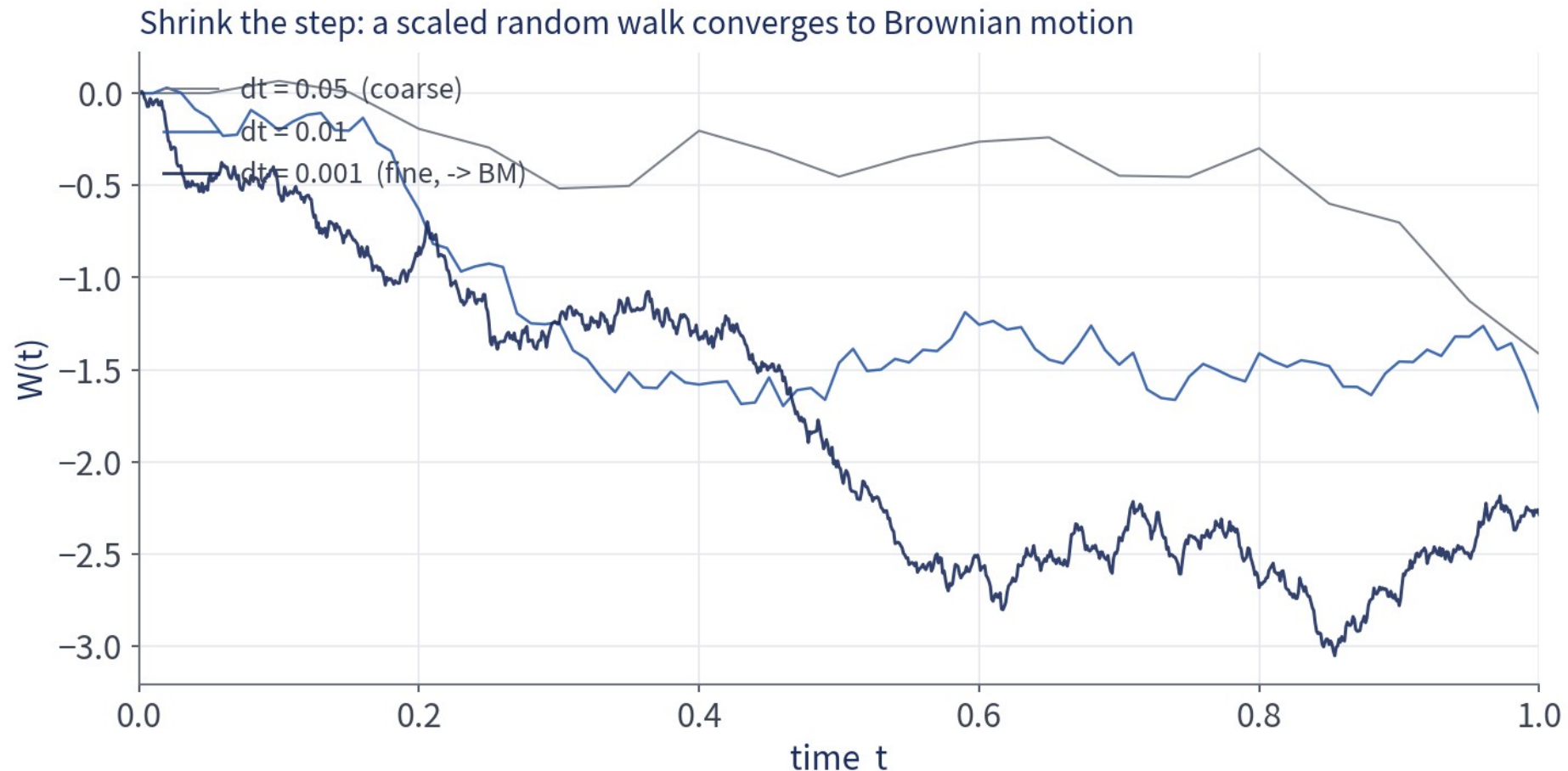
- An infinitesimal Brownian increment over dt .
- It is Gaussian with mean zero and variance dt .
- Its standard deviation scales as the square root of dt .

The key heuristic

- Squaring it gives dt , not a higher-order term.
- So (dW) squared cannot be discarded like (dt) squared.
- This single fact is what makes stochastic calculus different.

Random Walk Becomes Brownian Motion

Shrink the time step and a suitably scaled random walk fills in to a continuous Brownian path. Brownian motion is exactly this limit — the continuous-time cousin of the random walk.



Key Properties of Brownian Motion

Strange, but essential

What is true

- Continuous paths — no jumps.
- Independent, stationary Gaussian increments.
- Variance grows linearly with time.

What is surprising

- The paths are nowhere differentiable — infinitely jagged.
- So you cannot take an ordinary derivative dW/dt .
- That is why we work with increments dW , not slopes.

Stochastic Differential Equations

Trend plus randomness, in one line

$$dX_t = \underbrace{a(X_t, t) dt}_{\text{drift}} + \underbrace{b(X_t, t) dW_t}_{\text{diffusion}}$$

The two terms

- The drift a is the deterministic trend per unit time.
- The diffusion b scales the random Brownian shock.
- Both can depend on the current state and time.

Why this is general

- Almost every continuous-time model has this form.
- Choose a and b , and you choose the model.
- It is the continuous-time analogue of a recursion.

Reading an SDE

Drift steers, diffusion shakes

The drift term

- It is where the process is heading on average.
- Set it to zero and there is no trend.
- It captures growth, decay, or pull toward a level.

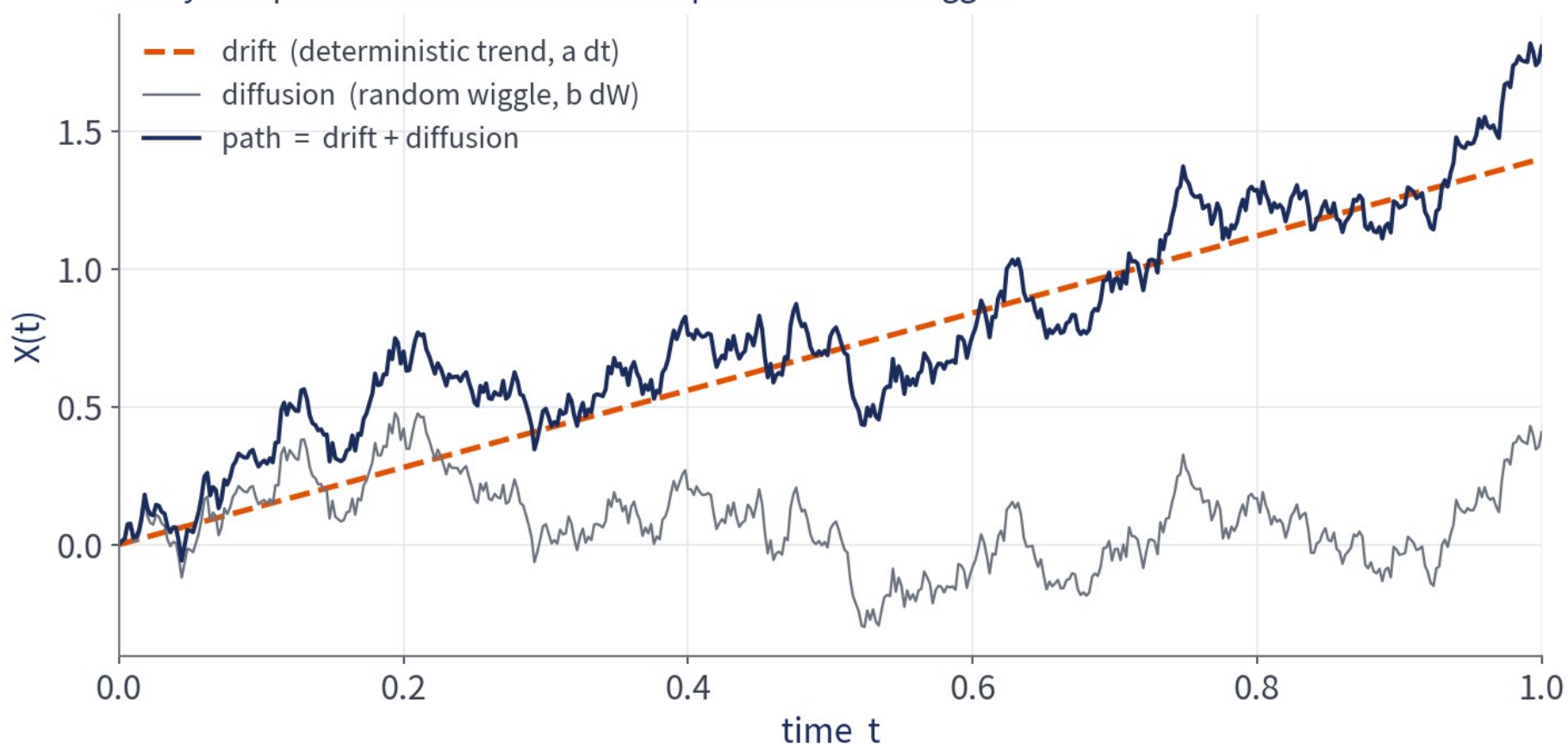
The diffusion term

- It is the size of the random fluctuations.
- Larger diffusion means a wilder path.
- It multiplies the Brownian increment dW .

Drift Plus Diffusion

Every SDE path splits into a smooth deterministic trend (drift) and a random wiggle around it (diffusion). Their sum is the realized path — trend plus noise, in continuous time.

Every SDE path is a deterministic trend plus a random wiggle



Geometric Brownian Motion

The standard model for an asset price

$$\text{GBM: } dS_t = \mu S_t dt + \sigma S_t dW_t$$

The specification

- Both drift and diffusion are proportional to the price.
- So returns, not prices, have constant statistics.
- The price stays positive and grows multiplicatively.

Why it fits prices

- Percentage moves, not dollar moves, are what's stable.
- It gives lognormally-distributed future prices.
- It is the engine behind the Black-Scholes formula.

Why GBM for Prices

Positive, proportional, tractable

Its virtues

- The price can never go negative.
- Moves scale with the level — a 1% move is a 1% move.
- It has clean closed-form solutions.

The intuition

- Model the log-price as Brownian motion with drift.
- Exponentiate to get a positive price.
- Simple, and good enough as a first approximation.

GBM's Strengths and Its Blind Spots

A great default, not a universal truth

What it gets right

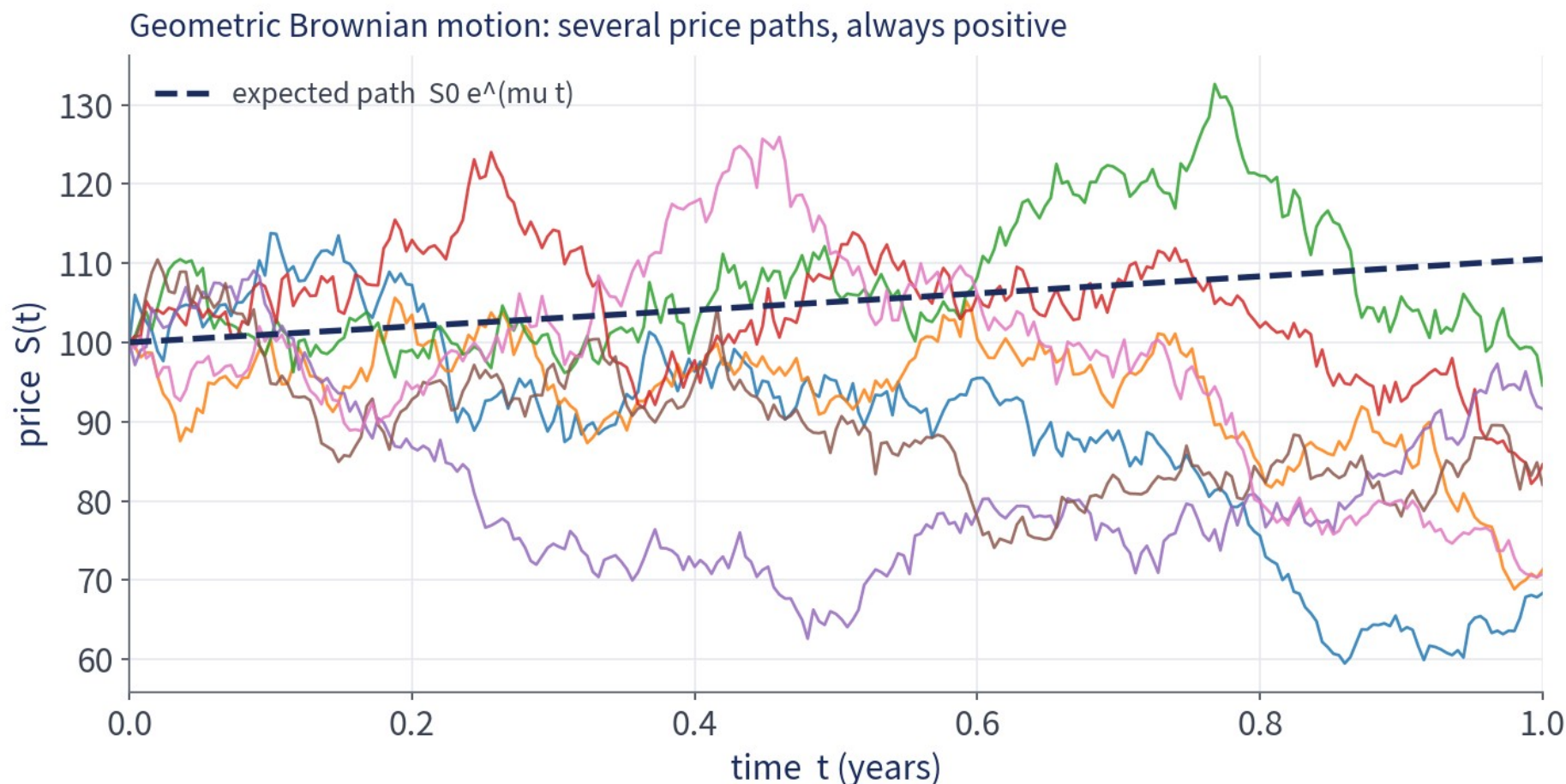
- Positivity and proportional moves.
- A tractable, well-understood benchmark.
- A sensible baseline for many assets.

What it misses

- Constant volatility — no clustering (recall GARCH).
- Thin (Gaussian) tails — no crashes.
- No mean reversion — prices wander without an anchor.

Geometric Brownian Motion in Action

Several GBM price paths from the same start: always positive, fanning out into a lognormal spread of outcomes around the expected exponential-growth path.



Ito's Lemma: The Stochastic Chain Rule

Differentiating a function of a random process

To work with functions of a random process — like the log of a price — you need a chain rule that accounts for the roughness of Brownian paths. That rule is Ito's lemma, and it carries an extra term.

The problem

- Ordinary calculus assumes smooth paths.
- Brownian paths are nowhere differentiable.
- So the usual chain rule does not apply.

The fix

- Ito's lemma extends the chain rule to SDEs.
- It adds a second-order term from the $(dW)^2 = dt$ fact.
- It is the workhorse of continuous-time finance.

The Extra Term

Where Ito departs from ordinary calculus

$$\text{Ito: } df = \left(f_t + a f_x + \frac{1}{2} b^2 f_{xx} \right) dt + b f_x dW_t \quad \leftarrow \quad \frac{1}{2} b^2 f_{xx} \text{ is the extra term}$$

The new piece

- Beyond the usual terms sits a second-derivative term.
- It is proportional to the diffusion squared.
- It comes directly from (dW) squared equalling dt .

Why it matters

- Ignore it and your dynamics are simply wrong.
- It is what makes log-GBM have a corrected drift.
- Every continuous-time derivation leans on it.

Why Ito Differs From Ordinary Calculus

Roughness leaves a footprint

Ordinary calculus

- Smooth paths: squared increments vanish faster than dt .
- So only first-order terms survive.
- The familiar chain rule holds.

Ito calculus

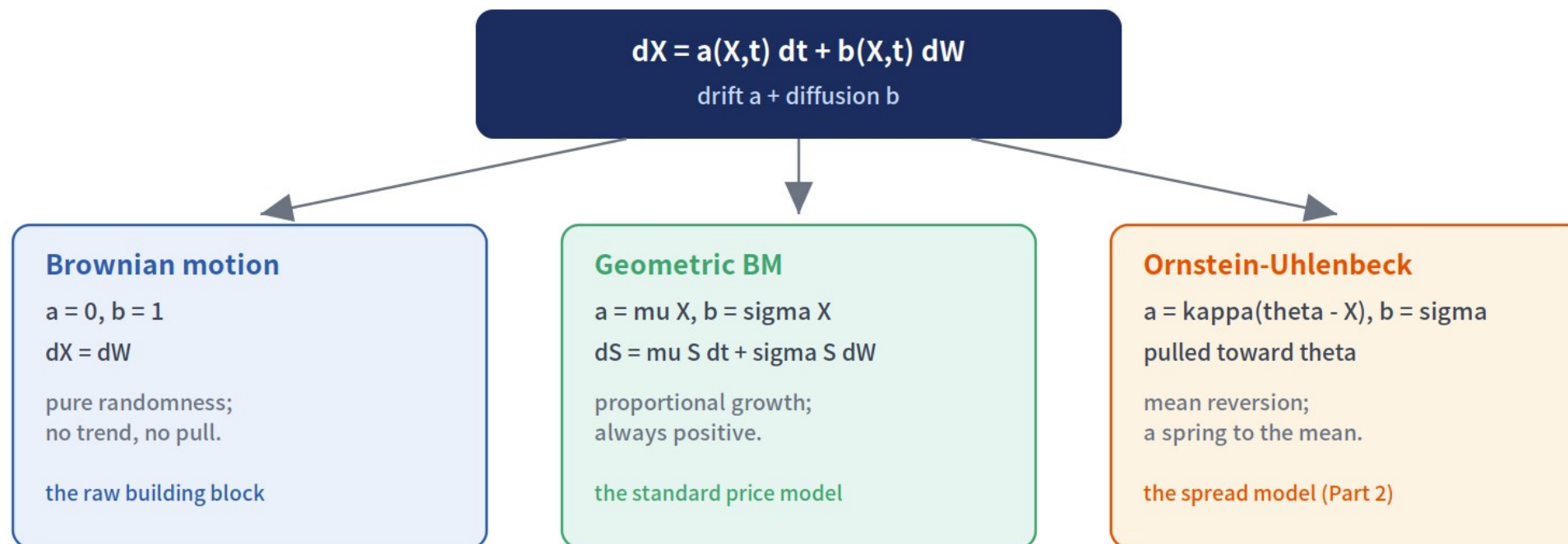
- Brownian squared increments are of order dt , not smaller.
- So a second-order term survives into the differential.
- That surviving term is the whole difference.

One Recipe, a Family of Models

Fix the drift and diffusion and the same SDE template becomes Brownian motion, geometric Brownian motion, or the Ornstein-Uhlenbeck process — the mean-reverting model we build next.

One Recipe, a Family of Models

Choose the drift and diffusion, and the same SDE template becomes a different model.



What Parts 2 and 3 Add

From randomness to mean reversion

Part 2 — the OU process

- An SDE with a drift that PULLS toward a mean.
- Continuous-time mean reversion, with three parameters.
- It is the continuous cousin of an AR(1).

Part 3 — in practice

- Model the pairs spread as an OU process (Stage V).
- Estimate its parameters and half-life.
- The cautions, and the bridge to interest-rate models.

Simulate Brownian Motion and GBM

Direct Claude Code to build the basics from scratch

GOAL Simulate Brownian motion and GBM, and see the random-walk-to-Brownian limit with your own eyes.

[INPUT] Prompt · Claude Code

Write and run `sde_basics.py` that:

1. simulates Brownian motion by cumulating Gaussian increments with variance dt ;
 2. shows that shrinking dt converges to a smooth Brownian path (overlay a few step sizes);
 3. simulates GBM $dS = \mu S dt + \sigma S dW$ via the exact log-Euler scheme, plotting paths;
 4. checks that the log-returns are Gaussian with the right mean and variance.
- Confirm GBM stays positive and returns look normal.

How to work with it

- Brownian variance should grow linearly with time.
- Smaller dt gives a finer, continuous-looking path.
- GBM paths must stay strictly positive.
- Keep it; Part 2 swaps in mean reversion.

Deliverable: `sde_basics.py` + Brownian and GBM paths, and a normality check on returns.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1

What is the variance of a Brownian increment over a time step dt ?

dt

Q2

In an SDE, what do the drift and diffusion terms each represent?

**Trend and
randomness**

Q3

Why does Ito's lemma carry an extra second-order term?

Because $(dW)^2 = dt$

All three come from the increment dW , the SDE recipe, and Ito's key fact.

Three Misconceptions to Avoid

Where continuous-time intuition slips

Common myths

- "Brownian paths have a well-defined velocity."
- "You can differentiate a function of a process as usual."
- "GBM captures crashes and volatility clustering."

The reality

- Brownian paths are nowhere differentiable — no velocity.
- Use Ito's lemma; it adds a second-order term.
- GBM has constant volatility and thin tails — no crashes.

Three Things to Carry Forward

Part 1 in one screen

1

Brownian motion is the building block

The continuous-time limit of a random walk: continuous, nowhere smooth, with Gaussian increments whose variance is dt . Its increment dW is the atom of stochastic calculus.

2

An SDE is drift plus diffusion

Choose a deterministic trend and a random diffusion and you specify a model. GBM — proportional drift and diffusion — is the standard, positive, lognormal price model.

3

Ito's lemma is the stochastic chain rule

Because $(dW)^2$ equals dt , differentiating a function of a process carries an extra second-order term. It is the workhorse of every continuous-time derivation.

Next — Part 2: the Ornstein-Uhlenbeck process, continuous-time mean reversion for a spread.

★ WEEK 11 · PART 1 · PRACTICE

Brownian Motion and SDEs, Worked by Hand

The Volatility Drag, and Why Risk Grows With Root-T

Ito's correction and Brownian scaling, made concrete

Part 1's continuous-time ideas, with the arithmetic shown. We apply Ito's lemma to see why a stock's median falls short of its mean, then use Brownian scaling to compute how volatility and VaR grow with the square root of time.

1P

PART 1 · APPLICATIONS

Two Continuous-Time Calculations

Case A: geometric Brownian motion and Ito's lemma - the drift correction that splits the mean from the median.

Case B: Brownian variance scaling - the square-root-of-time rule for volatility and VaR. Then a Claude Code lab.

What Part 1 Told Us

Now we make it concrete - with numbers

1

Brownian motion is the atom of randomness

Its variance grows with time, so its spread grows with the square root of time. That single fact drives risk scaling.

2

An SDE is drift plus diffusion

Geometric Brownian motion, $dS = \mu S dt + \sigma S dW$, is the standard asset-price model - drift steers, diffusion shakes.

3

Ito's lemma adds a correction

Differentiating a function of a random process leaves an extra half-sigma-squared term. On the log price, it's the volatility drag.

Everything here is a calculator exercise: one drift and vol, one daily volatility, two horizons.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from an SDE or a volatility.
- Apply Ito or the root-T rule by hand.
- Then read the financial consequence.

What you should watch

- How the mean and median part ways.
- How volatility scales with root-T.
- How naive linear scaling overstates risk.

The Volatility Drag

Why the median trails the mean

A high-growth stock has a 10% drift but a hefty 40% volatility. Its price follows geometric Brownian motion. Applying Ito's lemma to the log price reveals a correction - the typical outcome grows not at 10% but at 10% minus half the variance. We compute the gap between the mean and the median by hand

The inputs

- Drift $\mu = 10\%$ per year.
- Volatility $\sigma = 40\%$ per year.
- Start price 100.

What we'll do by hand

- Apply Ito to get the log-price drift.
- Compute the expected (mean) price.
- Compute the median, and the drag.

GBM and Ito on the Log Price

Where the correction comes from

GBM: $dS = \mu S dt + \sigma S dW$ (drift steers, diffusion shakes)

Ito on $\log S$: $d(\log S) = \left(\mu - \frac{1}{2}\sigma^2\right)dt + \sigma dW$

The price SDE

- Drift and diffusion both scale with S .
- That's what makes returns, not prices, stable.
- The standard model for an asset price.

The log transform

- Ito's lemma adds the -half-sigma-squared term.
- Ordinary calculus would miss it.
- It's the footprint of the roughness.

The Drift Correction

Ten percent becomes two

$$\text{expected log-return} = \mu - \frac{1}{2}\sigma^2 = 0.10 - \frac{1}{2}(0.40)^2 = 0.10 - 0.08 = \mathbf{0.02} \text{ (the Ito correction)}$$

The calculation

- Half of sigma-squared is half of 0.16.
- That's an 8% drag per year.
- Subtracted from the 10% drift.

The result

- The log price drifts at only 2% per year.
- This is the growth a typical path sees.
- Not the headline 10%.

Mean vs Median Price

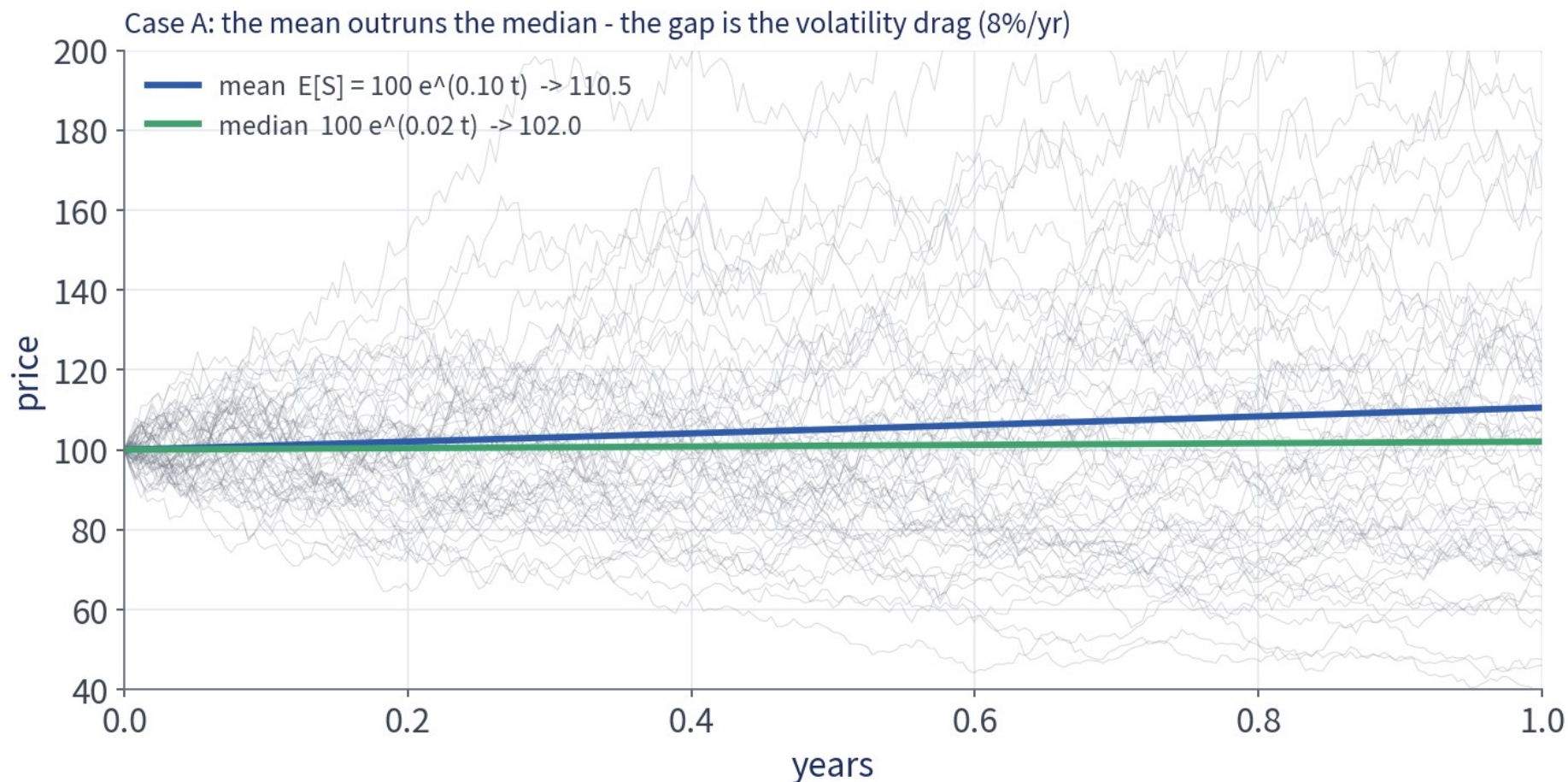
The same stock, two very different numbers

$$\begin{aligned}\text{mean: } \mathbb{E}[S_T] &= S_0 e^{\mu T} = 100 e^{0.10} = \mathbf{110.5} \\ \text{median: } S_0 e^{(\mu - \frac{1}{2}\sigma^2)T} &= 100 e^{0.02} = \mathbf{102.0}\end{aligned}$$

The average price after a year is 110.5, but the typical (median) price is only 102.0. The 8.5-point gap is the volatility drag eating into compound growth.

The Mean Outruns the Median

Simulated GBM paths fan out, but skew: a few big winners pull the mean (blue) up to 110.5, while the typical path - the median (green) - reaches only 102.0. The gap is the volatility drag, and it widens with volatility and time



Why Volatility Costs Growth

The drag is real money

The trap

- Quoting the average return as what you'll get.
- Ignoring that volatility eats compound growth.
- The higher the vol, the bigger the gap.

The reality

- A typical investor sees the median, not the mean.
- Half of sigma-squared is lost to the drag.
- Lower-vol compounding can beat higher-vol drift.

Risk Grows With Root-T

Brownian scaling, applied to VaR

Brownian motion's variance grows with time, so its standard deviation grows with the square root of time. That means a 10-day risk is not ten times a 1-day risk - it's about three times. We compute the 10-day volatility and VaR from a daily figure, and see why linear scaling badly overstates risk

The input

- Daily volatility of 1.5%.
- A 99% confidence VaR.
- Horizons of 1 and 10 days.

What we'll do by hand

- Scale volatility by root-10.
- Compute the 1-day and 10-day VaR.
- Compare to the naive linear answer.

Variance Scales With Time

So volatility scales with root-T

$$\text{Var}(W_t) = t \Rightarrow \text{vol over } T = \sigma\sqrt{T} \quad (\text{risk grows with } \sqrt{T}, \text{ not } T)$$

The property

- The variance of Brownian motion is the time.
- So its standard deviation is root-time.
- Independent increments add variances, not sds.

The consequence

- Volatility over T days is sigma times root-T.
- Risk grows sub-linearly with the horizon.
- The basis of the square-root-of-time rule.

The 10-Day Volatility

From daily to ten days

$$\sigma_{10} = \sigma_{\text{daily}} \sqrt{10} = 1.5\% \times 3.162 = \mathbf{4.74\%} \quad (\text{not } 1.5\% \times 10 = 15\%)$$

The calculation

- Multiply the daily 1.5% by root-10.
- Root-10 is about 3.16.
- That gives 4.74%, not 15%.

The intuition

- Ten days of shocks partly cancel.
- Variances add; volatilities don't.
- So the growth is root-10, not 10.

1-Day vs 10-Day VaR

Risk, scaled honestly

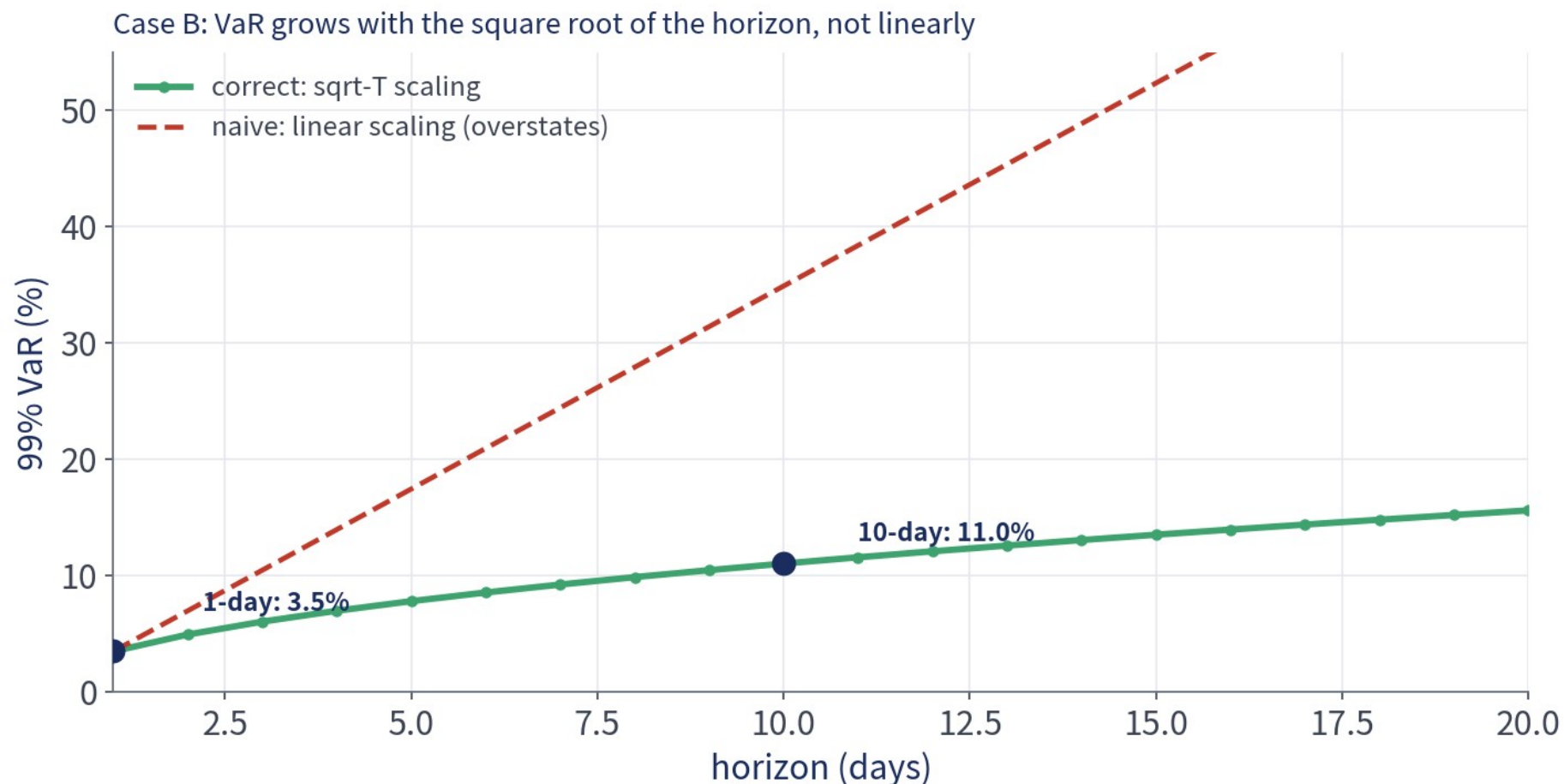
$$\text{1-day 99\% VaR} = 2.326 \times 1.5\% = 3.49\%$$

$$\text{10-day 99\% VaR} = 2.326 \times 1.5\% \times \sqrt{10} = \mathbf{11.0\%} \quad (\text{linear } \times 10 = 34.9\% \text{ is wrong})$$

The 10-day 99% VaR is 11.0%, not the 34.9% you'd get by scaling the 1-day figure linearly. Getting this right is the difference between a sensible and an absurd risk limit.

Root-T, Not Linear

The correct VaR (green) rises with the square root of the horizon, curving and flattening; the naive linear scaling (red) shoots up in a straight line and vastly overstates the risk at longer horizons. At 10 days the difference is 11.0% versus 25.0%



The Square-Root-of-Time Rule

And where it stops working

Why it holds

- Independent shocks add their variances.
- So volatility scales with root-T.
- It's the Basel rule for scaling VaR.

Where it breaks

- If returns are autocorrelated, root-T is off.
- Fat tails and jumps break the Gaussian scaling.
- It's an approximation, not a law.

The Continuous-Time Toolkit

Drift, diffusion, and a correction

Both cases flow from the same continuous-time picture. An SDE splits change into drift and diffusion; Brownian variance grows with time, giving root-T scaling; and Ito's lemma adds the half-sigma-squared correction whenever you transform the process. Get those three, and continuous-time finance opens up

The pieces

- Drift steers; diffusion shakes.
- Variance grows with time; vol with root-T.
- Ito adds the extra half-sigma-squared.

Why they matter

- The drag explains median vs mean.
- Root-T scales risk across horizons.
- Ito is the key to option pricing next.

Reproduce the Drag and the Scaling

Direct Claude Code to check your arithmetic

GOAL Verify the volatility drag and the root-T VaR, then simulate to confirm.

[INPUT] Prompt · Claude Code

Write and run `part1_practice.py` that:

CASE A - the drag:

- * from $\mu=0.10$, $\sigma=0.40$, $S_0=100$, print the log-drift $\mu-0.5\sigma^2$, the mean, and the median;
- * simulate GBM paths and check the mean and median.

CASE B - root-T:

- * from daily vol 1.5%, print the 10-day vol, the 1-day and 10-day 99% VaR, and the linear (x10) figure for contrast.

Match every printed number to the slides.

Check against the slides

- log-drift 0.02.
- mean 110.5, median 102.0.
- 10-day vol 4.74%.
- 10-day VaR 11.0% (linear 34.9%).

Deliverable: `part1_practice.py` whose printed numbers match the drag and the root-T scaling.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 What is the volatility drag?

**Half sigma-squared,
lost from compound
growth**

Q2 Why does the median trail the mean?

**Ito's correction on
the log price**

Q3 How does VaR scale with the horizon?

**With root-T, not
linearly**

All three come from Ito on the log price and Brownian variance scaling.

Three Pitfalls to Avoid

Where continuous time trips people

Common mistakes

- "The average return is what I'll compound at."
- "Differentiate log price like ordinary calculus."
- "Scale 1-day VaR to 10 days by multiplying by 10."

The reality

- The median grows at $\mu - \frac{1}{2}\sigma^2$.
- Ito adds the half-sigma-squared term - don't drop it.
- Scale by $\sqrt{10}$; linear scaling badly overstates risk.

Two Calculations to Remember

Part 1 applications in one screen

1

The volatility drag splits mean from median

Ito's lemma on the log price turns a 10% drift into a 2% typical growth rate. The mean price reaches 110.5 but the median only 102.0 - the 8%-per-year drag is half of sigma-squared, lost to volatility.

2

Risk grows with the square root of time

Brownian variance scales with time, so the 10-day volatility is 1.5% times root-10 = 4.74%, and the 10-day 99% VaR is 11.0% - not the 34.9% linear scaling wrongly gives. Variances add, not volatilities.

3

Drift, diffusion, and Ito's correction

An SDE separates trend from noise, Brownian motion scales with root-T, and Ito adds the half-sigma-squared term on any transform. These three unlock continuous-time finance - and option pricing ahead.

Next — Part 2 practice: the Ornstein-Uhlenbeck process, mean reversion in continuous time.

★ WEEK 11 · PART 2 of 3

The Ornstein-Uhlenbeck Process

Mean Reversion, in Continuous Time

The spring mechanism · κ , θ , σ · half-life · a continuous AR(1)

Brownian motion wanders without an anchor. The Ornstein-Uhlenbeck process adds a drift that pulls the state back toward a mean — the continuous-time model of a mean-reverting spread.

2

PART 2 · THE OU PROCESS

A Spring That Pulls to the Mean

The OU stochastic differential equation, the reversion mechanism, the roles of its three parameters, the half-life, and the exact discretization that makes OU a continuous-time AR(1).

From Wandering to Reverting

Part 1 built SDEs; Part 2 adds a pull

1

We have the SDE template

Drift plus diffusion. Brownian motion and GBM both wander — nothing pulls them back to a level.

2

But spreads revert

A cointegrated pairs spread doesn't drift away; it is tethered to a mean. We need an SDE that captures that pull.

3

The OU process is that model

An SDE whose drift points back toward a mean — continuous-time mean reversion, and the continuous cousin of the AR(1).

Part 2 in one line: add a mean-reverting drift and Brownian motion becomes the OU process.

The Ornstein-Uhlenbeck Process

An SDE with a pull toward the mean

$$\text{OU: } dX_t = \underbrace{\kappa(\theta - X_t)}_{\text{pull to the mean}} dt + \sigma dW_t$$

The drift

- It is proportional to the gap between the mean and X .
- When X is above the mean, the drift points down.
- When X is below, it points up — always toward the mean.

The diffusion

- A constant-size Brownian shock, as before.
- It pushes X around at random.
- The drift fights it, pulling X back home.

From GBM to OU

One wanders, the other is tethered

GBM (Part 1)

- The drift pushes in one direction, forever.
- The process wanders with no anchor.
- Good for prices, which trend and compound.

OU (this part)

- The drift reverses sign across the mean.
- The process is pulled back whenever it strays.
- Good for spreads, which revert to an equilibrium.

The Reversion Mechanism

A spring pulling toward the mean

Think of the state attached to the mean by a spring. Stretch it away and the spring pulls back — harder the farther you stretch. That restoring force is exactly the OU drift, κ times the gap.

The spring analogy

- The mean is the spring's resting position.
- Displacement creates a restoring force toward it.
- Bigger displacement, stronger pull — proportional.

In the SDE

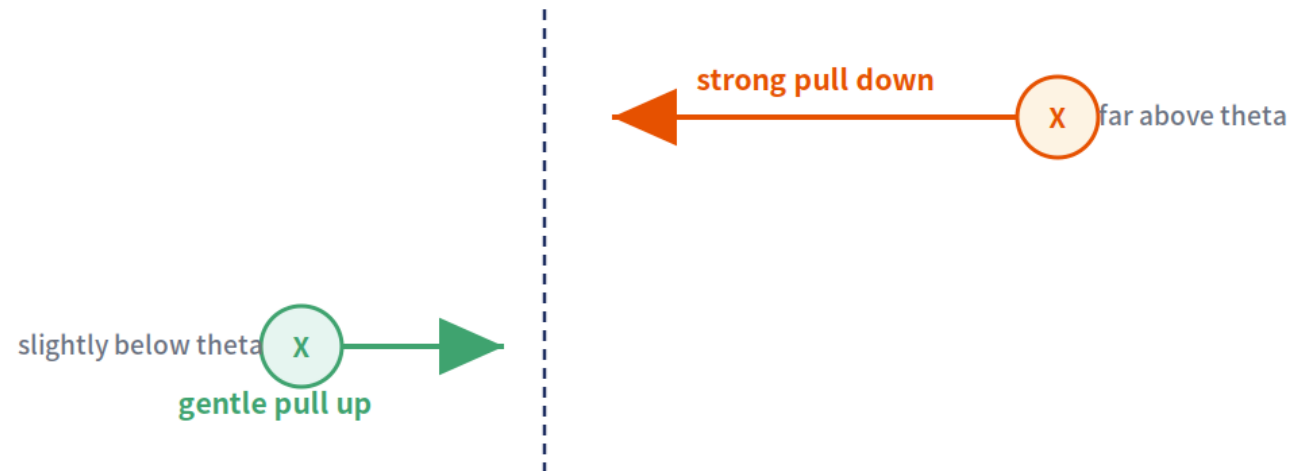
- The restoring force is the drift, κ times (mean minus X).
- The noise keeps knocking X away from the mean.
- The two balance into fluctuations around the mean.

The Reversion Mechanism: a Spring

The drift always points toward the mean and grows with distance. Far above, a strong pull down; slightly below, a gentle pull up. The gap sets direction; the speed kappa sets strength.

The Reversion Mechanism: a Spring

The drift always points back toward theta, and pulls harder the farther X strays.



$$\text{drift} = \text{kappa} (\text{theta} - X)$$

The gap ($\text{theta} - X$) sets the direction; kappa sets the strength. Bigger gap, stronger restoring force - exactly like a spring. The noise σdW pushes X around, but the spring keeps dragging it home.

The Three OU Parameters

What kappa, theta, and sigma control

Parameter	Role	What it controls
kappa	reversion speed	how fast X is pulled back to the mean
theta	long-run mean	the level X reverts toward
sigma	volatility	the size of the random shocks

Three numbers fully describe an OU process: where it settles, how strongly, and how noisily.

What kappa Controls

The speed of the pull

Large kappa

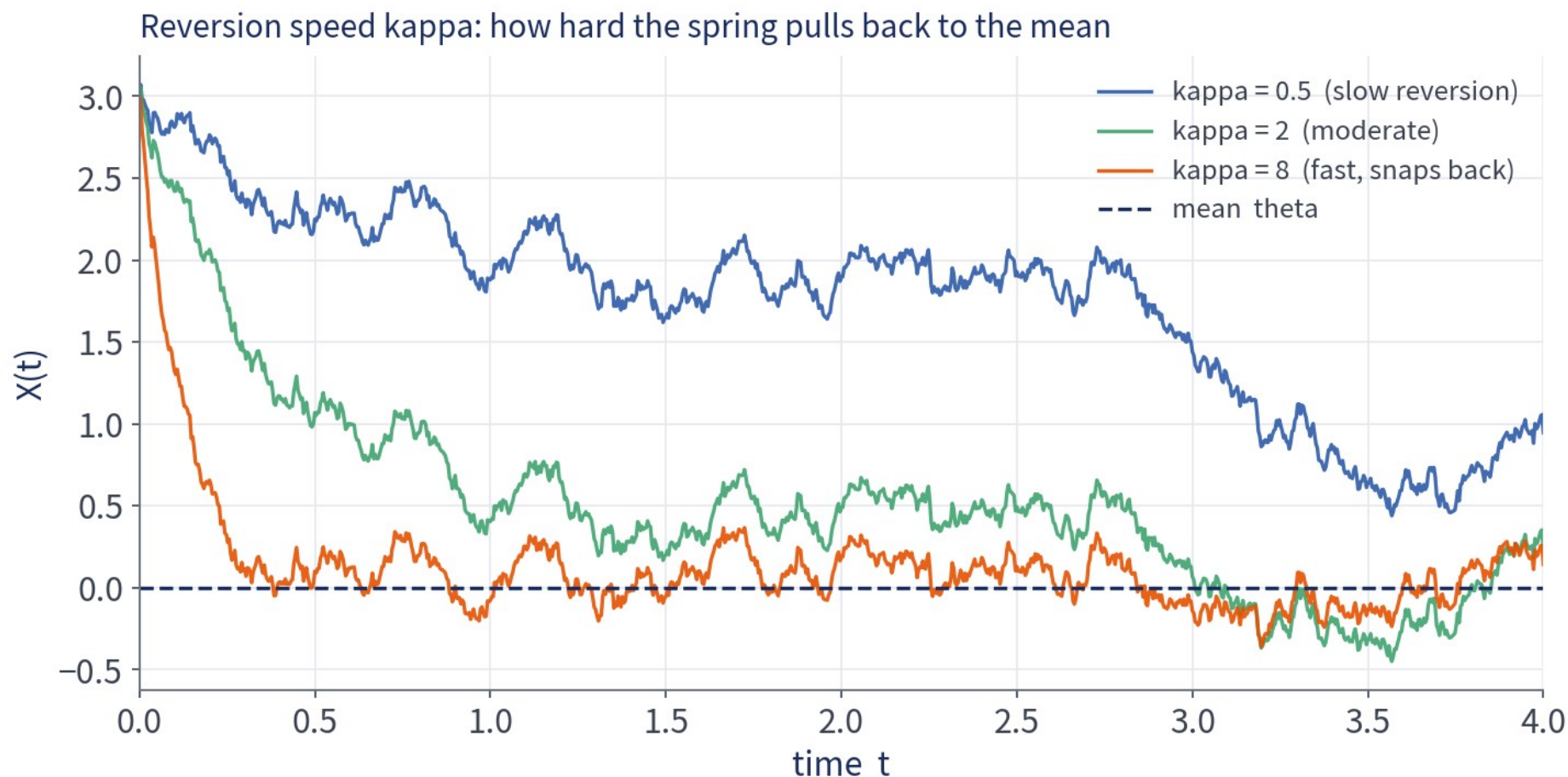
- A strong, fast spring.
- X snaps back to the mean quickly.
- Short excursions, short half-life.

Small kappa

- A weak, slow spring.
- X drifts far and lingers before returning.
- Long excursions, long half-life.

Reversion Speed in Pictures

Three OU paths starting away from the mean, differing only in kappa. A large kappa snaps back fast; a small one wanders and returns slowly. The mean and noise are identical.



The Stationary Distribution

Where OU settles over time

$$\text{stationary law: } X_{\infty} \sim \mathcal{N}\left(\theta, \frac{\sigma^2}{2\kappa}\right)$$

The long-run law

- Left alone, OU settles into a Gaussian distribution.
- Its mean is theta — the reversion level.
- Its variance balances noise against reversion speed.

Reading the variance

- More noise (sigma) widens the spread.
- Faster reversion (kappa) tightens it.
- The spread is sigma-squared over twice kappa.

OU Half-Life

How long to close half the gap

$$\text{half-life: } t_{1/2} = \frac{\ln 2}{\kappa} \quad (\text{time to close half the gap})$$

What it is

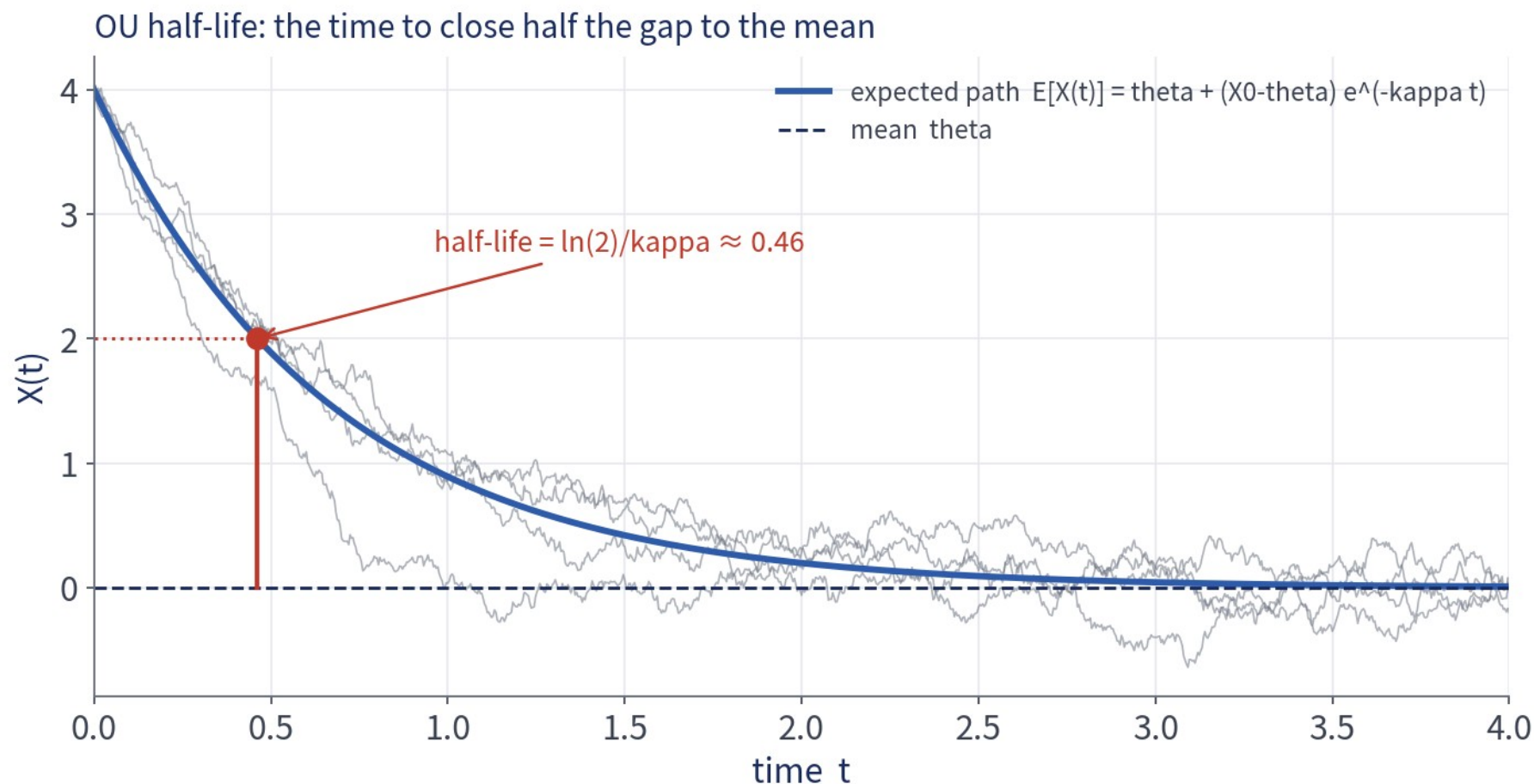
- The time for the expected gap to the mean to halve.
- It is the natural of the reversion speed.
- Short half-life means quick mean reversion.

Why traders care

- It sets the horizon of a mean-reversion trade.
- It tells you how long to expect to hold.
- We first met it as a discrete idea in the pairs thread.

The Half-Life, Illustrated

Starting away from the mean, the expected path decays exponentially back toward it. The half-life marks where half the initial gap has closed — the natural clock of the process.



The OU Process in Action

Pulled back, unlike Brownian motion

The defining behavior: OU stays near its mean. Shocks push it away, but the spring always drags it back. Over time it fluctuates in a band around the mean, rather than wandering off like Brownian motion.

What you see

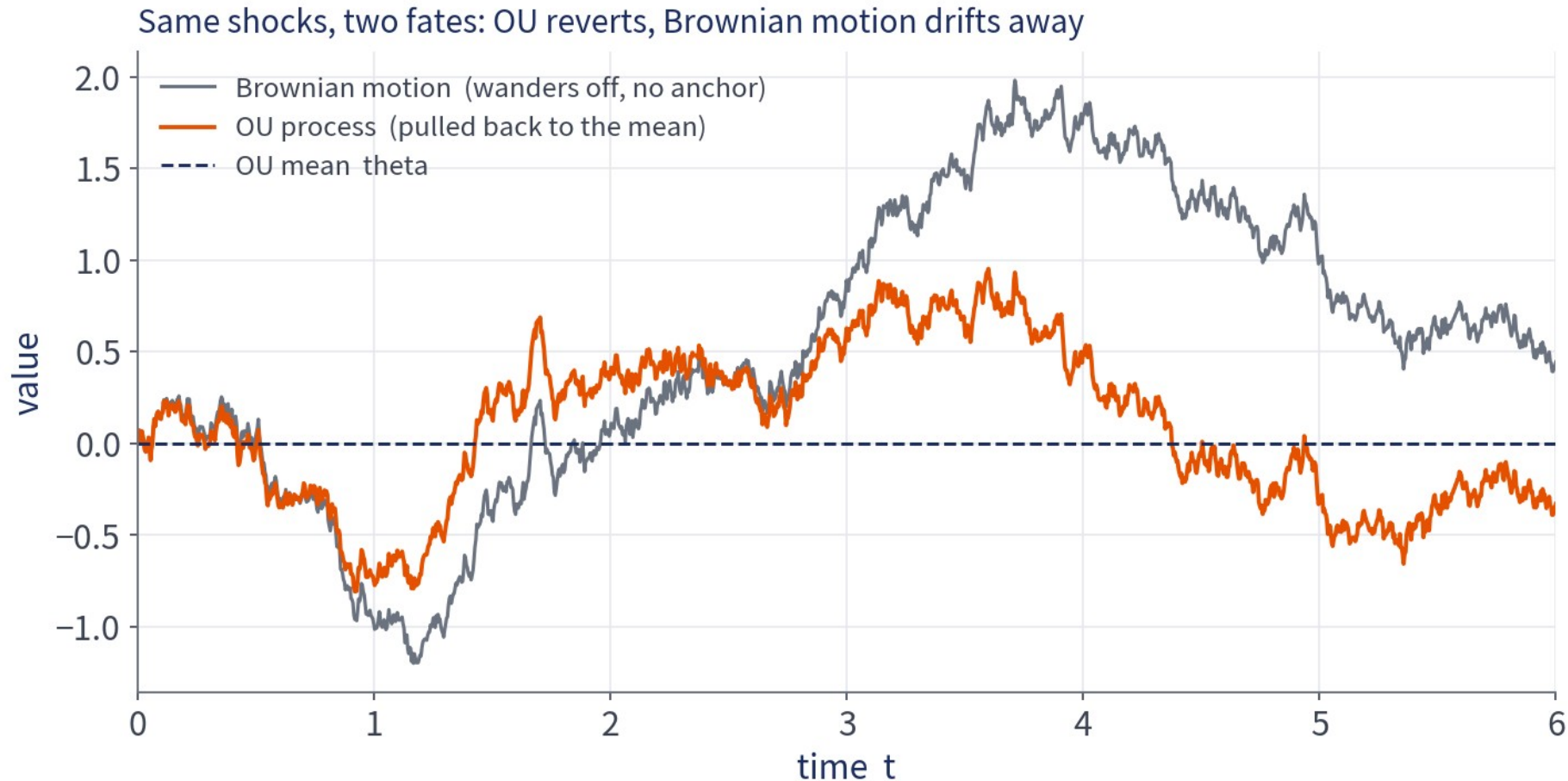
- A path that hovers around a central level.
- Excursions that are pulled back, not sustained.
- A stable, bounded spread of values.

Why it fits spreads

- A cointegrated spread behaves exactly this way.
- It is tethered to an equilibrium, not free to drift.
- OU gives a clean model of that tether.

OU vs Brownian Motion

Driven by the very same shocks, the OU process (orange) is pulled back to its mean while Brownian motion (grey) wanders off. The difference is entirely the mean-reverting drift.



Discretization: OU Is a Continuous AR(1)

The bridge from this week back to Part I

Sample the OU process at fixed intervals and you get exactly an AR(1) — the very model from the start of the course. Continuous-time mean reversion and discrete-time autoregression are the same idea.

The correspondence

- Observing OU on a grid gives an AR(1) recursion.
- The persistence ϕ is set by the reversion speed.
- The mean and noise carry over directly.

Why it is satisfying

- Two whole frameworks turn out to describe one process.
- The continuous view gives intuition (a spring).
- The discrete view gives an easy estimation route.

The Exact Discretization

OU sampled on a grid

$$X_{t+\Delta t} = \theta + e^{-\kappa\Delta t} (X_t - \theta) + \varepsilon_t \quad \varepsilon_t \sim \mathcal{N}\left(0, \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa\Delta t})\right)$$

Reading it

- Next value = mean + a decayed gap + Gaussian noise.
- The decay factor is the exponential of minus kappa times dt.
- This is exactly the AR(1) form, with a mean.

It is exact, not approximate

- No Euler error — the OU SDE solves in closed form.
- The formula holds for any step size dt.
- So you can estimate it cleanly from sampled data.

OU Is a Continuous-Time AR(1)

Discretize the OU SDE and the AR(1) from Part I reappears: the persistence ϕ equals the exponential of minus κ times dt . Estimate the AR(1), then read off the continuous parameters.

OU Is a Continuous-Time AR(1)

Discretize the OU SDE and you recover the AR(1) from Part I - the bridge back to discrete time.

Continuous time (OU)

$$dX = \kappa(\theta - X) dt + \sigma dW$$

κ - reversion speed

θ - long-run mean

σ - volatility

$$\text{half-life} = \ln(2) / \kappa$$

discretize

step size dt

Discrete time (AR(1))

$$X_{t+1} = \theta + \phi(X_t - \theta) + \epsilon$$

$\phi = e^{(-\kappa dt)}$ - persistence

same θ - the mean

ϵ - Gaussian innovation

$$\kappa = -\ln(\phi) / dt$$

Estimate the AR(1) in discrete time (easy), then read off the continuous OU parameters - reversion speed and half-life - from ϕ . Two languages for one mean-reverting process.

The Parameter Mapping

From AR(1) persistence to reversion speed

$$\text{it is an AR(1): } \phi = e^{-\kappa \Delta t} \iff \kappa = -\frac{\ln \phi}{\Delta t}$$

One key relation

- Persistence ϕ is the exponential of minus κ times Δt .
- So κ is minus the log of ϕ , divided by Δt .
- High persistence (ϕ near 1) means slow reversion.

What it lets you do

- Fit an AR(1) — a one-line regression.
- Convert ϕ into κ and a half-life.
- Report continuous-time quantities from discrete data.

Why This Bridge Matters

Estimate discretely, interpret continuously

The practical route

- Real data arrive on a grid — daily, hourly.
- AR(1) estimation on that grid is trivial.
- The mapping then hands you the OU parameters.

The conceptual gain

- The continuous model gives the spring intuition and half-life.
- The discrete model gives a clean estimator.
- You get the best of both descriptions.

What Part 3 Adds

From model to trade

Pairs Stage V

- Model the cointegrated spread as an OU process.
- Read off its mean, half-life, and typical range.
- Turn those into principled entry and exit rules.

Estimation and cautions

- How to fit OU parameters from a spread series.
- Why fitting is easy but trusting is hard.
- The bridge to interest-rate models (Week 12).

Simulate and Estimate an OU Process

Direct Claude Code to build OU intuition

GOAL Simulate an OU process, then recover its parameters via the AR(1) bridge — and check the half-life.

[INPUT] Prompt · Claude Code

Write and run `ou_estimate.py` that:

1. simulates an OU process for known κ , θ , σ (exact discretization);
2. estimates it by regressing X_{t+1} on X_t (an AR(1)) and recovering κ , θ , σ ;
3. converts the AR(1) ϕ into κ via $\kappa = -\ln(\phi)/dt$, and the half-life;
4. plots the path with the mean and the half-life horizon marked.

Confirm the recovered parameters match the truth.

How to work with it

- ϕ near 1 means slow reversion.
- Recovered κ should match the input.
- Half-life = $\ln(2)/\kappa$.
- Part 3 applies this to a real spread.

Deliverable: `ou_estimate.py` + recovered κ , θ , σ , and a half-life estimate.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1 What does the OU drift do when X is above the mean?

Pulls it down

Q2 What is the OU half-life in terms of the reversion speed?

$\ln(2) / \kappa$

Q3 Sampled on a grid, an OU process is exactly what model?

An AR(1)

All three come from the reversion drift, the half-life formula, and the discretization.

Three Misconceptions to Avoid

Where OU intuition slips

Common myths

- "OU and Brownian motion look basically the same."
- "A larger sigma means slower reversion."
- "Discretizing OU is only an approximation."

The reality

- OU is tethered to a mean; Brownian motion wanders off.
- Reversion speed is κ ; sigma only sets the noise size.
- OU's discretization is EXACT — a true AR(1), any Δt .

OU at a Glance

Everything on one card

Quantity	Formula / form	Meaning
Dynamics	$dX = \kappa(\theta - X)dt + \sigma dW$	the OU SDE
Long-run mean	θ	where it settles
Long-run variance	$\sigma^2 / (2\kappa)$	spread around the mean
Half-life	$\ln(2) / \kappa$	time to close half the gap
Discrete form	AR(1), $\phi = e^{(-\kappa dt)}$	the estimation bridge

Keep this card: it is the whole of the OU process in five lines.

Three Things to Carry Forward

Part 2 in one screen

1

OU is Brownian motion with a spring

Its drift, κ times the gap to the mean, pulls the state back toward θ whenever it strays. The noise pushes; the spring pulls; the two balance into fluctuations around the mean.

2

Three parameters, one half-life

Reversion speed κ , mean θ , and volatility σ fully describe it. The half-life, $\log_2 2 / \kappa$, is the natural clock of the reversion — and the horizon of a trade.

3

OU is a continuous-time AR(1)

Sampled on a grid it is exactly an AR(1) with persistence $e^{-\kappa \Delta t}$. Estimate the AR(1), then read off the continuous parameters — the bridge back to Part I.

Next — Part 3: model the pairs spread as an OU process (Stage V), estimate it, and trade it.

★ WEEK 11 · PART 2 · PRACTICE

The OU Process, Worked by Hand

Fit Mean Reversion, Then Trade It

From an AR(1) coefficient to a half-life and a spread trade

Part 2's mean-reversion model, with the arithmetic shown. We fit an Ornstein-Uhlenbeck process to a spread through its AR(1) coefficient - reading off a half-life and equilibrium band - then turn those into entry, exit, and an expected holding time.

2P

PART 2 · APPLICATIONS

Estimate Discretely, Interpret Continuously

Case A: fit OU to a spread via its AR(1) coefficient - mapping to the reversion speed, half-life, and equilibrium band.
Case B: trade that spread - entry and exit from the band, holding time from the half-life. Then a Claude Code lab.

What Part 2 Told Us

Now we fit and trade it - with numbers

1

OU is Brownian motion with a leash

The drift pulls the process back toward a mean θ at speed κ . Unlike GBM, it doesn't wander off - it reverts.

2

The half-life measures the pull

How long to close half the gap to the mean: $\ln 2$ over κ . It's the natural timescale of the reversion.

3

OU is a continuous AR(1)

Fit an AR(1) to the discrete data, map its coefficient to κ , and interpret the result in continuous time.

Everything here is a calculator exercise: one AR(1) coefficient, one half-life, one band.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a fitted AR(1) coefficient.
- Map to kappa, half-life, and the band.
- Then turn those into a trade.

What you should watch

- How the coefficient becomes a reversion speed.
- How the half-life sets the timescale.
- How the band sets entry and exit.

Fit OU to a Spread

From a coefficient to a continuous model

A pairs spread reverts to zero. We fit a simple AR(1) to the daily data and get a coefficient of 0.90. Because OU is a continuous AR(1), that one number maps to the reversion speed kappa, from which we read the half-life and the equilibrium band - a full continuous-time characterization estimated discretely

The inputs

- Daily AR(1) coefficient $\phi = 0.90$.
- Innovation $sd = 0.30$, mean $\theta = 0$.
- Sampling interval one day.

What we'll do by hand

- Map ϕ to the reversion speed κ .
- Compute the half-life.
- Compute the equilibrium band.

OU and Its AR(1) Bridge

One process, two representations

$$\begin{aligned} \text{OU: } dX &= \kappa(\theta - X) dt + \sigma dW && \text{(a pull toward } \theta) \\ \text{discretized: } X_t &= \phi X_{t-1} + \dots, \quad \phi = e^{-\kappa \Delta t} && \text{(OU is a continuous AR(1))} \end{aligned}$$

In continuous time

- The drift pulls toward theta at speed kappa.
- Sigma is the size of the shocks.
- That's the OU stochastic differential equation.

On a grid

- Sampled daily, OU becomes an AR(1).
- Its coefficient is e-to-the-minus-kappa.
- So the coefficient carries all the reversion.

Speed and Half-Life

From the coefficient to the timescale

$$\kappa = -\frac{\ln \phi}{\Delta t} = -\ln 0.90 = 0.105/\text{day}, \quad \text{half-life} = \frac{\ln 2}{\kappa} = \frac{0.693}{0.105} = \mathbf{6.6} \text{ days}$$

The mapping

- kappa is minus the log of the coefficient.
- A coefficient of 0.90 gives kappa 0.105.
- Higher persistence, slower reversion.

The half-life

- $\ln 2$ divided by kappa is 6.6 days.
- That's the time to close half the gap.
- The natural clock of the spread.

The Equilibrium Band

Where the spread lives

$$\text{stationary sd} = \frac{\sigma_\varepsilon}{\sqrt{1-\phi^2}} = \frac{0.30}{\sqrt{1-0.81}} = \mathbf{0.69}, \quad \text{band } \pm 2\text{sd} = \pm 1.38$$

The stationary sd

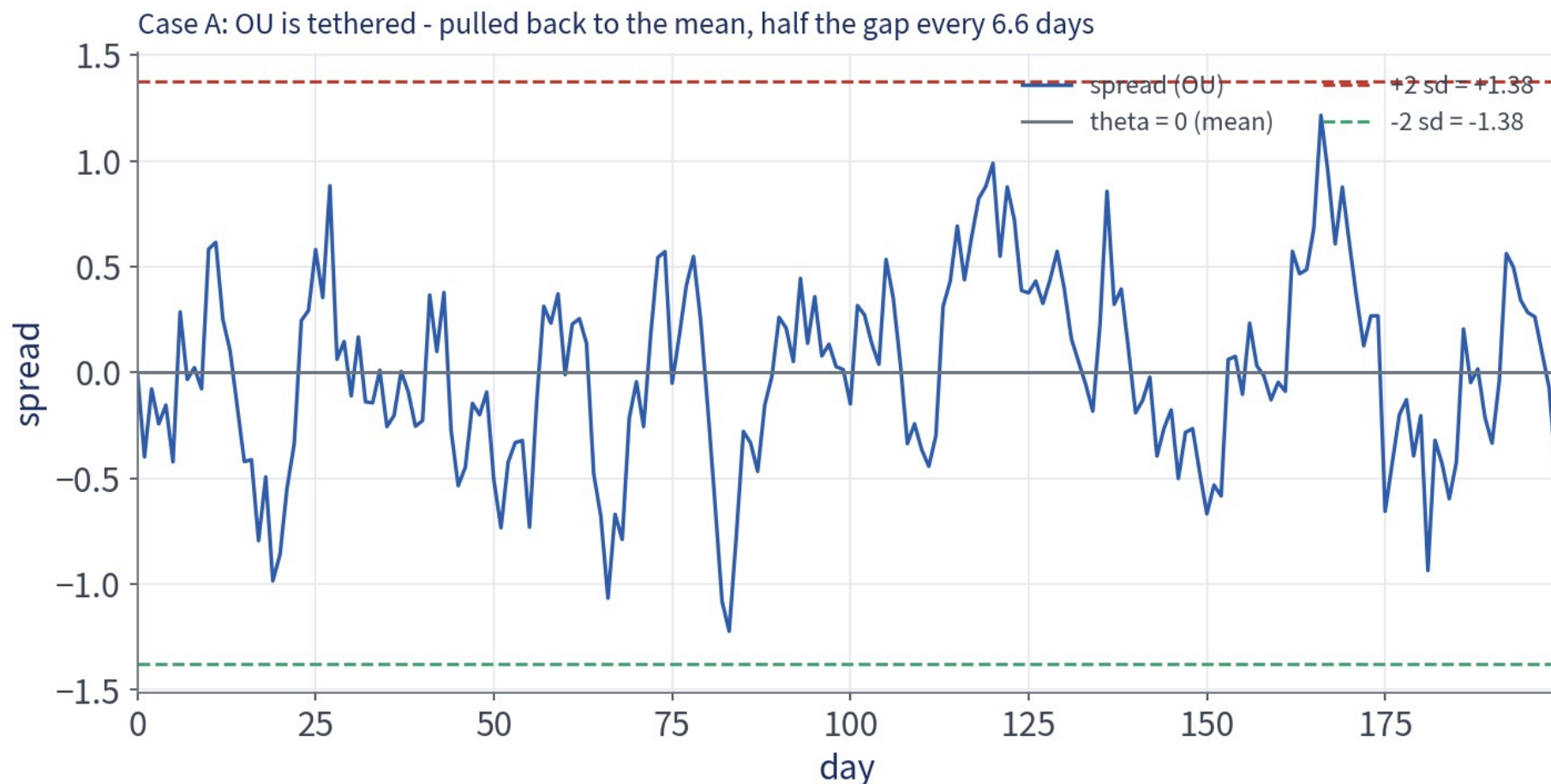
- Innovation sd over root of one-minus-phi-squared.
- That's 0.30 over root 0.19, about 0.69.
- It's how far the spread typically strays.

The band

- Two standard deviations is plus or minus 1.38.
- The spread sits inside this band most of the time.
- It becomes the trigger for a trade.

Tethered, Not Wandering

Unlike Brownian motion, the OU spread is pulled back whenever it strays - closing half the gap to the mean every 6.6 days. It spends most of its time inside the plus-or-minus-1.38 band, which is exactly what makes it tradable



Estimate Discretely, Interpret Continuously

The bridge that makes OU practical

Why the bridge matters

- You never observe continuous time - only samples.
- AR(1) is trivial to fit to the daily data.
- The mapping recovers the continuous parameters.

What you get

- A reversion speed with real-world units.
- A half-life you can act on.
- An equilibrium band for trading.

Trade the OU Spread

Entry, exit, and holding time. **With OU fitted, the trade writes itself. Enter when the spread reaches the edge of its band - two standard deviations from the mean - betting it reverts. The expected gap then decays exponentially, halving every half-life, so the holding time follows directly. We compute the reversion path and the expected hold by**

The plan

- Enter at plus or minus 1.38.
- Exit as the spread nears the mean.
- The half-life sets the timescale.

What we'll do by hand

- Set the entry and exit thresholds.
- Track the expected gap decaying.
- Read off the expected holding time.

Entry and Exit From the Band

The band becomes the trigger

enter at $|X| \geq 2 \text{sd} = 1.38$ (short if high, long if low), exit near $X = \theta = 0$

The entry

- When the spread hits two sd, it's stretched.
- Short it if high, buy it if low.
- Bet on the pull back to the mean.

The exit

- Close as the spread returns near theta.
- The reversion is the profit.
- The band sizes both ends of the trade.

The Reversion, Halving

How the gap closes over time

$$\mathbb{E}[X_t] = X_0 e^{-\kappa t} \Rightarrow \text{the gap halves every half-life : } 2 \text{ sd} \rightarrow 1 \text{ sd} \rightarrow 0.5 \text{ sd}$$

The decay

- The expected gap follows e-to-the-minus-kappa-t.
- Each half-life halves what remains.
- From 2 sd to 1 sd to 0.5 sd.

Reading it

- One half-life: down to one sd.
- Two: down to half a sd.
- The pull is fast at first, then eases.

Expected Holding and Profit

How long, and how much

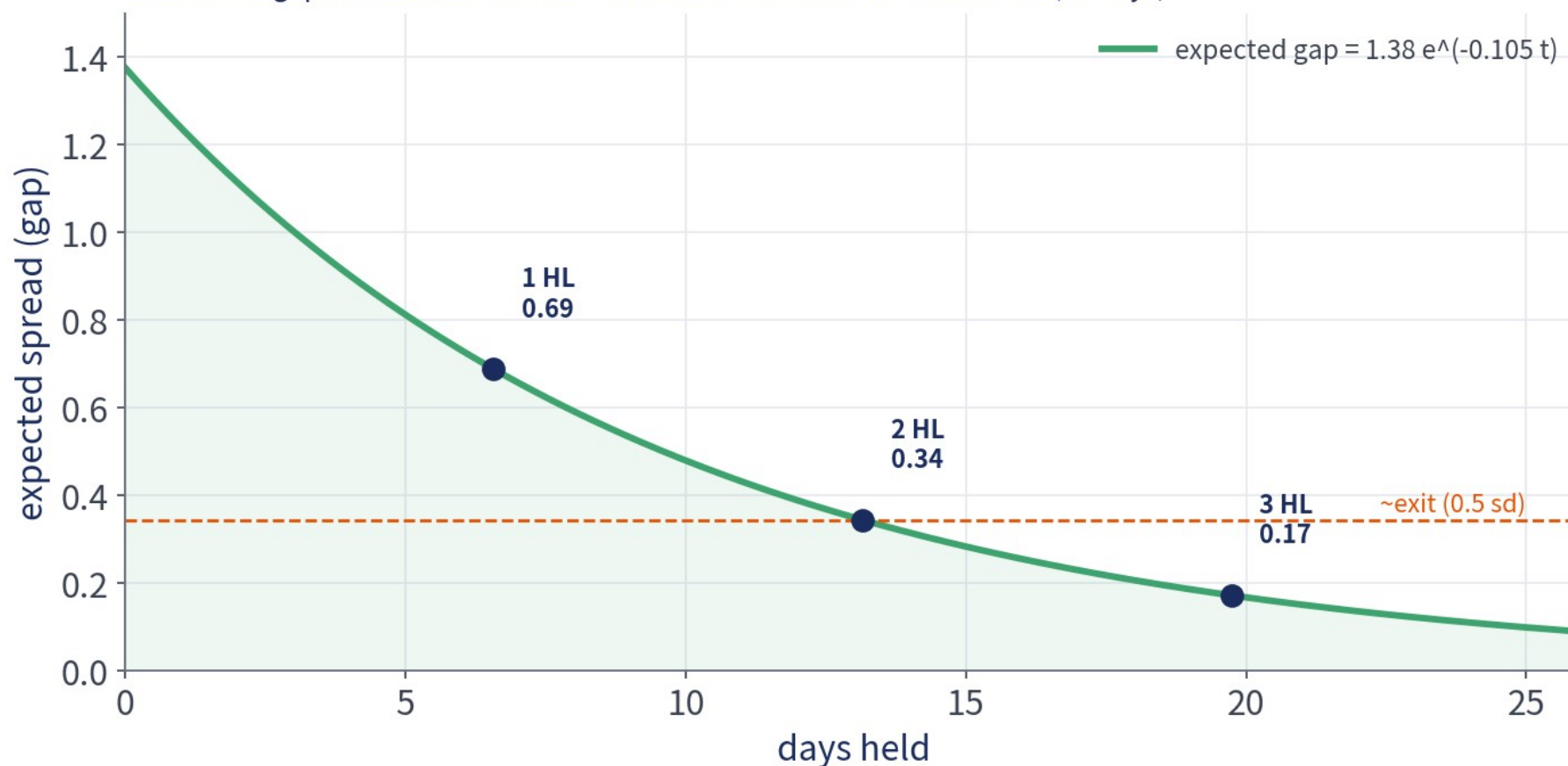
from 2 sd to $\sim 0.25\text{ sd}$: ≈ 3 half-lives = **20** days (expected holding)
profit per trade $\approx$ captured reversion = 1.38 spread units (before costs)

From two sd back toward the mean takes about three half-lives - some 20 days. The captured reversion, about 1.38 spread units, is the gross profit before costs.

The Gap Halves Each Half-Life

Enter at the top of the band (1.38) and the expected spread decays exponentially: one half-life brings it to 1 sd, two to 0.5 sd, three to about 0.25 sd - near the exit. The half-life is the clock that tells you how long your capital is committed

Case B: the gap halves each half-life - back near the mean in ~3 half-lives (20 days)



The Half-Life Runs the Trade

Timescale and thresholds from one fit

What the fit gives you

- The band tells you when to enter.
- The half-life tells you how long to wait.
- Both from a single AR(1) coefficient.

Why it's disciplined

- A slow reverter ties up capital longer.
- A wide band means bigger, rarer trades.
- The parameters set the whole trade profile.

Tethered Randomness, Made Tradable

Fit, then trade. Both cases turn on one idea: OU is mean reversion you can measure. Fit an AR(1), map it to a reversion speed, and you have a half-life and an equilibrium band. Those two numbers are the entire trade - when to enter, how far the move is worth, and how long you'll wait for it. Estimate discretely, interpret and trade

The fit (Case A)

- AR(1) coefficient to reversion speed.
- Half-life of 6.6 days.
- Equilibrium band of plus or minus 1.38.

The trade (Case B)

- Enter at the band, exit at the mean.
- Gap halves every half-life.
- Held about 20 days for 1.38 units.

Reproduce the Fit and the Trade

Direct Claude Code to check your arithmetic

GOAL Verify the OU fit and the reversion decay, then backtest the spread trade.

[INPUT] Prompt · Claude Code

Write and run `part2_practice.py` that:

CASE A - fit OU:

- * from AR(1) $\phi=0.90$, innovation $sd=0.30$, $dt=1$,
print $\kappa=-\ln(\phi)$, the half-life, the
stationary sd , and the ± 2 sd band;

CASE B - trade it:

- * set entry at 2 sd , exit at the mean;
- * print the expected gap at 1, 2, 3 half-lives;
- * report the expected holding time.

Then fit OU to a real spread and backtest the trade.

Match every printed number to the slides.

Check against the slides

- $\kappa = 0.105$ per day.
- half-life = 6.6 days.
- stationary $sd = 0.69$, band ± 1.38 .
- hold ~ 20 days.

Deliverable: `part2_practice.py` whose printed numbers match the OU fit and the reversion decay.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How do you get κ from an AR(1) coefficient?

**κ = minus the
log of the coefficient**

Q2 What is the half-life?

**$\ln 2$ over κ - time
to close half the gap**

Q3 How does the expected gap decay over time?

**Exponentially -
halving every half-
life**

All three come from the AR(1)-to-OU mapping and the reversion decay.

Three Pitfalls to Avoid

Where OU trading slips

Common mistakes

- "Read kappa straight off the coefficient."
- "Assume the spread reverts on a fixed schedule."
- "Trust the fit when the relationship has changed."

The reality

- kappa is minus the log of the coefficient, not one minus it.
- Reversion is exponential in expectation, not a countdown.
- Re-estimate - a broken spread has no half-life.

Two Calculations to Remember

Part 2 applications in one screen

1

Fit OU through its AR(1) coefficient

A coefficient of 0.90 maps to a reversion speed κ of 0.105 per day, a half-life of 6.6 days, and a stationary sd of 0.69 - an equilibrium band of plus or minus 1.38. A full continuous model, estimated from daily data.

2

Trade it from the band and the half-life

Enter at two sd (1.38) and the expected gap halves every half-life: 2 sd to 1 sd to 0.5 sd. Reverting near the mean takes about three half-lives - some 20 days - to capture roughly 1.38 spread units.

3

One fit sets the whole trade

The band says when to enter, the reversion says how far, and the half-life says how long. Estimate discretely, interpret and trade continuously - the OU bridge in one workflow.

Next — Part 3 practice: OU in trading and Stage V, where mean reversion completes the pairs strategy.

★ WEEK 11 · PART 3 of 3

OU in Practice: Pairs Trading Stage V

A Principled Mean-Reversion Model of the Spread

OU model of the spread · estimation · half-life filter · the cautions

The OU process finally gives the pairs spread a model, not just thresholds: a fair value, statistically-grounded bands, and a half-life. We estimate it, trade it, and confront what can go wrong.

3

PART 3 · IN PRACTICE

An OU Model of the Spread

Pairs trading Stage V models the spread as an OU process, estimated through the AR(1) bridge, with the half-life as a trade filter — plus the cautions, and the road to interest-rate models.

From the Model to the Trade

Parts 1-2 built OU; Part 3 uses it

1

We have a mean-reversion model

The OU process, with a spring toward the mean, three parameters, and a half-life — and it discretizes to an AR(1).

2

The pairs spread reverts

A cointegrated spread is exactly the kind of mean-reverting series OU was made for. So model it as OU.

3

Stage V makes the trade principled

Fair value, entry bands, and a horizon all fall out of the fitted OU parameters — replacing ad-hoc thresholds with an estimable model.

Part 3 in one line: model the spread as OU, estimate it, and let its parameters set the trade.

Stage V: An OU Model of the Spread

The thread gets a principled mean-reversion model

Earlier stages built the spread and protected it. Stage V finally MODELS it. Treating the spread as an OU process gives a fair value, bands grounded in its statistics, and a half-life — all estimable from data.

What we had

- A validated, mean-reverting spread (Stage II).
- An adaptive hedge and a regime filter (III, IV).
- But entry rules were still z-score thresholds.

What Stage V adds

- A model: the spread IS an OU process.
- Fair value, bands, and horizon from its parameters.
- A principled, estimable basis for the trade.

Why Model the Spread as OU

From thresholds to a model

The problem with thresholds

- A z-score of two is an arbitrary line.
- It says nothing about how fast reversion happens.
- It gives no horizon and no fair value estimate.

What OU provides

- A fair value: the reversion level θ .
- Bands from the spread's own long-run spread.
- A half-life that sets the expected holding time.

The Spread as an OU Process

One SDE for the whole trade

model the spread as OU: $d(\text{spread}_t) = \kappa(\theta - \text{spread}_t) dt + \sigma dW_t$

Reading it

- The spread is pulled toward theta at speed kappa.
- Theta is the fair value — where it belongs.
- Sigma sets how much it wanders around theta.

Why it fits

- A cointegrated spread reverts, exactly as OU does.
- The tether from Week 7 becomes a spring here.
- The model is simple, and its parameters are estimable.

From OU to Trading Bands

Bands grounded in the spread's statistics

$$\text{bands: enter when } |\text{spread} - \theta| > z \sigma_{eq}, \quad \sigma_{eq} = \frac{\sigma}{\sqrt{2\kappa}}$$

The entry band

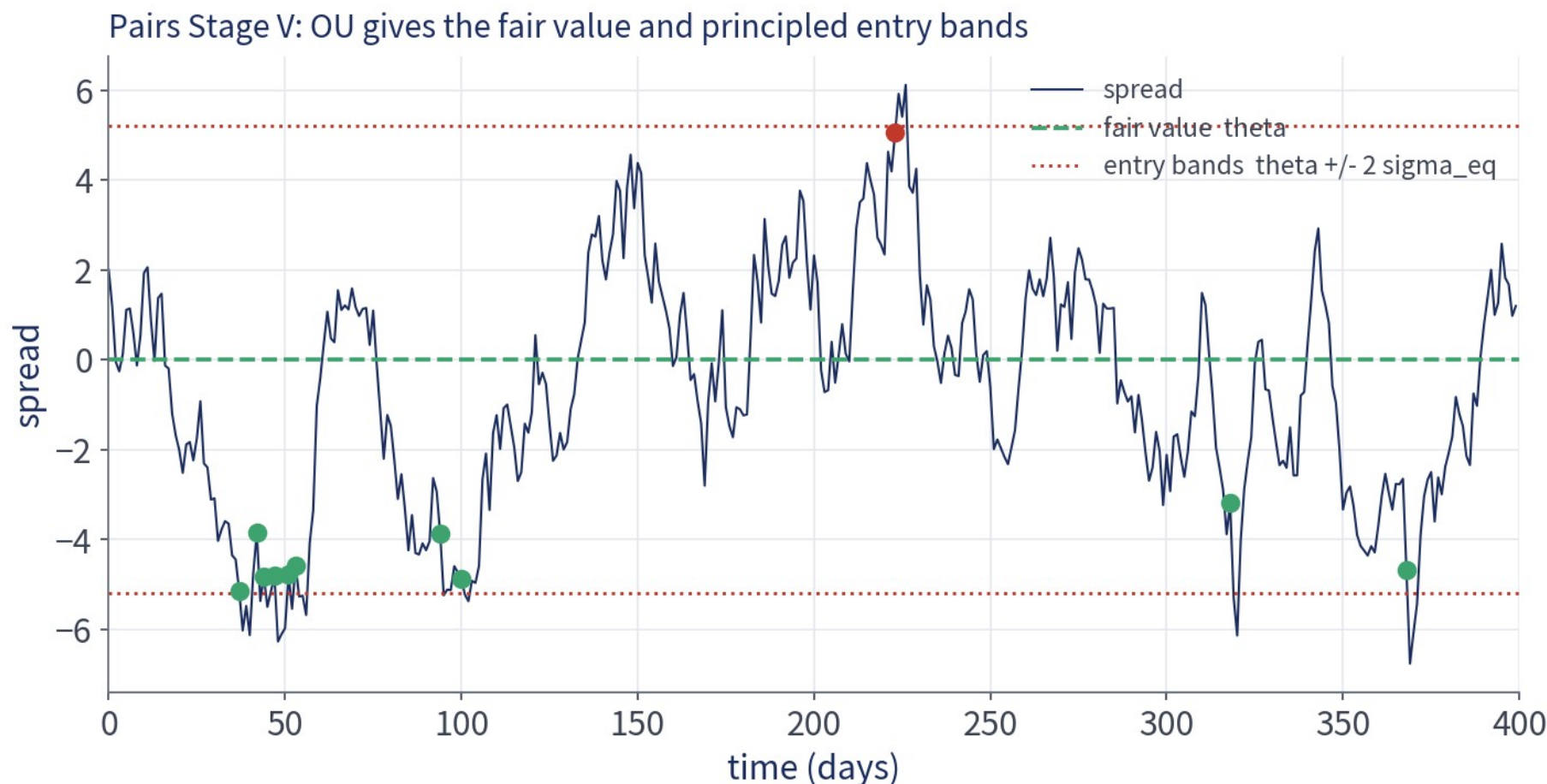
- Enter when the spread is far from theta ...
- ... measured in units of its long-run spread.
- That spread is sigma over the root of twice kappa.

The exit

- Exit as the spread reverts back toward theta.
- The half-life tells you roughly how long that takes.
- Bands and horizon both come from the fit.

Stage V: OU Bands on the Spread

The fitted OU gives a fair value (theta) and entry bands set by the spread's stationary spread. Fade the spread when it crosses a band and exit as it reverts — a model-based mean-reversion trade.



What OU Adds Over Earlier Stages

Principle where there were heuristics

Before Stage V

- Bands were fixed z-scores, chosen by hand.
- No sense of how long a trade should last.
- No explicit estimate of fair value.

With Stage V

- Bands scale with the spread's own statistics.
- The half-life gives a natural holding horizon.
- Theta is an explicit, estimated fair value.

Estimating the OU Parameters

Three numbers from a spread series

Fitting OU means recovering kappa, theta, and sigma from a spread. The cleanest route uses the AR(1) bridge from Part 2: regress the spread on its lag, then translate the persistence into a reversion speed and half-life.

The routes

- AR(1) regression — the discretization bridge.
- Maximum likelihood on the OU transition density.
- They agree; the regression is the simplest.

What you get out

- Theta from the regression's implied mean.
- Kappa from the persistence coefficient.
- Sigma and the half-life follow directly.

Estimation via the AR(1) Bridge

A one-line regression, then translate

$$\text{estimate via AR(1): } \phi \Rightarrow \kappa = -\frac{\ln \phi}{\Delta t}, \quad \theta, \quad t_{1/2} = \frac{\ln 2}{\kappa}$$

The steps

- Regress the spread on its own one-step lag.
- The slope is the AR(1) persistence ϕ .
- Convert ϕ into κ , and thus a half-life.

Why it's convenient

- It is a single ordinary regression.
- It uses the exact OU-to-AR(1) mapping.
- Fast enough to run across many candidate pairs.

The Half-Life as a Trade Filter

Screen pairs by how fast they revert

Too fast

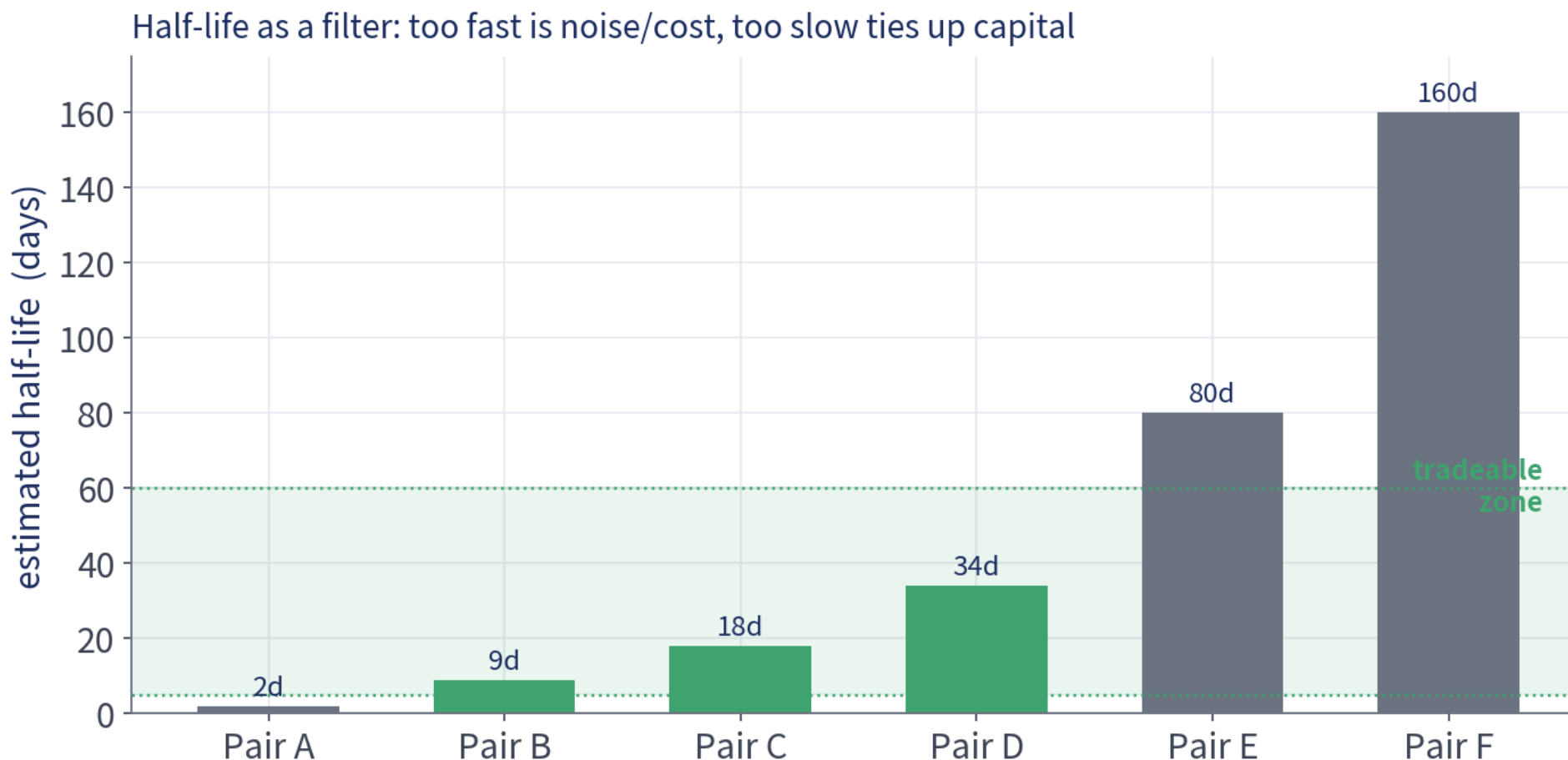
- A very short half-life reverts within noise.
- Trading costs eat any edge.
- Often it is just microstructure, not signal.

Too slow

- A very long half-life ties up capital for ages.
- Reversion may not complete before things change.
- Prefer a sensible, tradeable middle range.

Half-Life as a Filter

Estimated half-lives across candidate pairs. Too short is noise and cost; too long ties up capital. A tradeable zone (shaded) keeps only pairs that revert on a usable horizon.



Estimation and Cautions

Fitting OU is easy; trusting it is the hard part

You can always fit an OU to a spread — the regression never fails. The hard part is trusting the fit. If the spread isn't truly stationary, or its parameters drift, the tidy fair value and bands can be dangerously misleading.

The easy part

- The regression always returns some κ , θ , σ .
- You get a fair value and bands automatically.
- It all looks clean and authoritative.

The hard part

- Is the spread actually stationary right now?
- Are the parameters stable, or drifting?
- A fit on a broken spread is worse than no model.

Non-Stationarity and Regime Breaks

When the spread stops reverting

What goes wrong

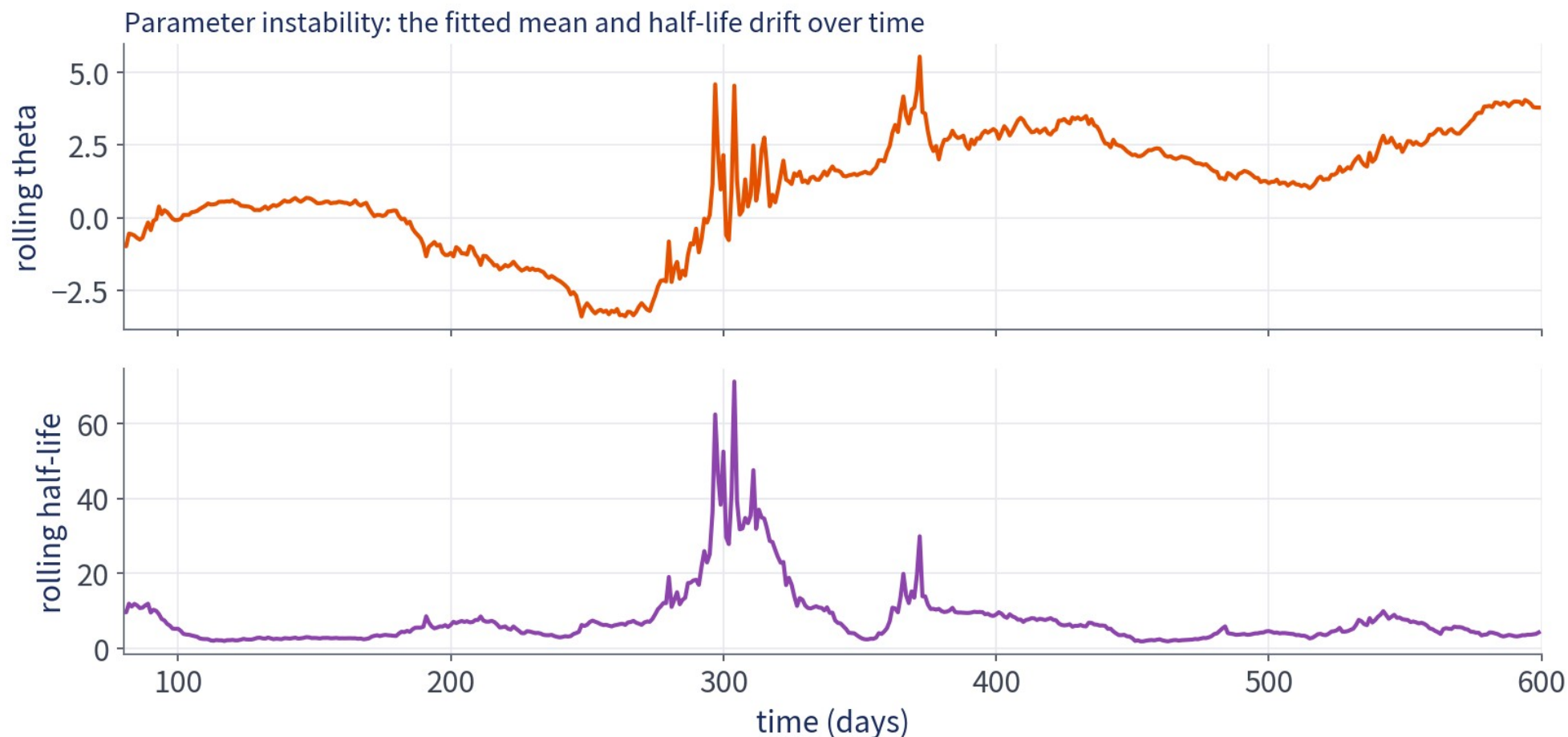
- The cointegration can break (Week 7).
- In a crisis the spread may trend, not revert (Week 10).
- OU then models an equilibrium that no longer exists.

How to guard

- Re-test stationarity, don't assume it persists.
- Overlay the regime filter from Stage IV.
- Stand down when the mean-reversion premise fails.

Parameters Drift Over Time

Re-estimating OU on a rolling window shows the fair value and half-life wandering as conditions change. A single static fit hides this — treat the parameters as moving targets, not constants.



Living With Drifting Parameters

Re-estimate, and stay humble

Practical habits

- Re-estimate the OU parameters on a rolling window.
- Watch the half-life; a lengthening one warns of trouble.
- Size positions down as uncertainty rises.

The mindset

- Treat θ and κ as slowly moving, not fixed.
- Combine OU with the earlier stages, not in isolation.
- The model informs the trade; it does not command it.

The Pairs Journey, Now Five Stages

Volatility bands (W5), a cointegration anchor (W7), a Kalman hedge (W9), a regime filter (W10), and now an OU model (W11) that turns the spread into a fair value, bands, and a half-life.

The Pairs Journey, Now Five Stages

Five weeks, five layers - Stage V gives the spread a principled continuous-time model.



What Stage V contributes

Earlier stages built and protected the spread; Stage V finally MODELS it. The OU process gives a fair value (theta), statistically-grounded entry bands (from the stationary spread), and a half-life that sets the horizon.

From ad-hoc thresholds to a principled, estimable model of mean reversion.

The Continuous-Time Toolkit

Everything Week 11 gave you

Object	Form	Use
Brownian motion	dW	raw continuous randomness
Geometric BM	$dS = \mu S dt + \sigma S dW$	asset prices (positive, lognormal)
Ornstein-Uhlenbeck	$dX = \kappa(\theta - X)dt + \sigma dW$	mean-reverting spreads
Ito's lemma	adds a second-order term	functions of a process

Four objects, one language: continuous-time models built on Brownian motion.

When OU Is the Right Model

Match the model to the series

OU fits when

- The series genuinely reverts to a stable mean.
- The reversion is roughly linear in the gap.
- Shocks are roughly Gaussian and constant in size.

OU is wrong when

- The series trends or has a unit root (use GBM/ARIMA).
- Reversion speed or mean shifts sharply (add regimes).
- Jumps or fat tails dominate (need richer models).

Trade the Spread as an OU Process

Direct Claude Code to build the Stage V trade

GOAL Fit an OU model to a pairs spread, derive bands and a half-life, and back-test the mean-reversion trade.

[INPUT] Prompt · Claude Code

Write and run `pairs_stage5.py` that:

1. builds a cointegrated spread and fits OU via the AR(1) regression (recover κ , θ , σ , and the half-life);
2. sets fair value θ and entry bands at $\theta \pm z * \sigma_{eq}$ ($\sigma_{eq} = \sigma / \sqrt{2 \kappa}$);
3. back-tests: enter beyond a band, exit near θ ;
4. re-estimates OU on a rolling window and plots how θ and the half-life drift.

Report the half-life and whether the pair is tradeable, and flag when parameters look unstable.

How to work with it

- Bands should come from the fit, not a guess.
- Half-life sets the holding horizon.
- Watch rolling θ and κ for drift.
- Combine with Stage IV regimes in crises.

Deliverable: `pairs_stage5.py` + OU fit, model-based bands, a back-test, and a drift check.

Three Questions Before You Finish

If you can answer these, Week 11 has landed

Q1 In Stage V, what plays the role of fair value?

The OU mean theta

Q2 How do you estimate OU from a spread most simply?

An AR(1) regression

Q3 Why is fitting OU easy but trusting it hard?

Stationarity can fail

All three come from the OU spread model, the AR(1) estimator, and the stationarity caution.

Three Pitfalls to Avoid

Where Stage V goes wrong

Common mistakes

- "A clean OU fit means the trade is safe."
- "The fitted θ and κ are constants."
- "Any half-life is tradeable."

The reality

- A fit on a broken spread is worse than none.
- Parameters drift — re-estimate and watch them.
- Screen out half-lives that are too short or too long.

Week 11 in Three Ideas

Pulling the whole week together

1

Continuous time, built on Brownian motion

Parts 1-2: Brownian motion and the increment dW , SDEs as drift plus diffusion, GBM for prices, Ito's lemma, and the OU process — continuous-time mean reversion with a spring toward the mean.

2

OU is a continuous-time AR(1)

Its half-life is log-two over the reversion speed, and sampling it on a grid gives an AR(1). So you estimate it as a simple regression and read off the continuous parameters.

3

Stage V models the spread

The pairs spread becomes an OU process: a fair value, statistically-grounded bands, and a half-life filter. Fitting is easy; trusting it — checking stationarity and drift — is the real work.

Continuous time gave the spread a principled model — and a language for what comes next.

How It All Connects

Eleven weeks, one thread

The through-line

- AR(1) reversion (W2) -> cointegration (W7) -> Kalman (W9).
- Regimes (W10) -> a continuous-time OU model (W11).
- Each week added a layer to the same pairs trade.

The recurring ideas

- Stationarity, mean reversion, and hidden states.
- Estimate discretely, interpret through a model.
- Every tool is judged by whether it survives real data.

Where We Go From Here

Week 12 · Interest-Rate Models and Discretization

1

Interest-rate models are OU's cousins

The classic short-rate models (Vasicek and its relatives) are mean-reverting SDEs — the OU process, applied to interest rates. Week 11 is the foundation for all of them.

2

Discretization becomes central

To simulate and estimate these SDEs, you discretize them. The exact OU discretization and the Euler scheme are the tools we sharpen next.

3

From one spread to whole curves

We move from modeling a single mean-reverting spread to modeling interest rates and the term structure — continuous-time finance in full.

See you in Week 12 — mean-reverting SDEs for interest rates, and how to discretize them.

★ WEEK 11 · PART 3 · PRACTICE

OU in Trading, Worked by Hand

Screen Pairs by Half-Life; Turn Reversion Into Return

Stage V - the economics of a mean-reversion portfolio

Part 3's Stage V, with the arithmetic shown. We use the half-life as a filter to screen candidate pairs, then compute the trade economics - how a faster reverter recycles capital into more annual return from the same per-trade edge.

3P

PART 3 · APPLICATIONS

From One Pair to a Portfolio

Case A: screen four candidate pairs by half-life, keeping only fast, stationary reverters. Case B: the trade economics - how the half-life drives turnover and annual return. Then the five-stage journey and a Claude Code lab.

What Part 3 Told Us

Now we screen and size it - with numbers

1

OU replaces heuristics with a model

Instead of ad-hoc thresholds, the spread's own statistics set the bands - principle where there was guesswork.

2

The half-life is a trade filter

It tells you how fast a pair reverts, so you can screen candidates and keep only the ones worth trading.

3

Re-estimate and stay humble

Spreads drift and break. A fitted half-life is only as good as the relationship that produced it.

Everything here is a calculator exercise: four candidate pairs, one filter, two economics.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from fitted AR(1) coefficients.
- Screen by half-life or size the trade.
- Then read the portfolio decision.

What you should watch

- How the filter rejects slow and non-reverting pairs.
- How the half-life drives trades per year.
- How turnover turns the same edge into more return.

Screen Pairs by Half-Life

Not every spread is worth trading

You have four candidate pairs, each with a fitted AR(1) coefficient. The half-life turns each into a single, comparable number - how fast it reverts. A good trade needs a half-life short enough to turn capital, but the pair must actually revert. We screen all four by hand

The filter

- Accept a half-life between 2 and 30 days.
- Too slow ties up capital for little return.
- The coefficient must be below one to revert.

What we'll do by hand

- Convert each coefficient to a half-life.
- Apply the filter to all four.
- Keep the tradeable ones.

The Half-Life Filter

One number to screen a pair

$$\text{half-life} = \frac{\ln 2}{-\ln \phi}, \quad \text{ACCEPT if } 2 \leq \text{HL} \leq 30 \text{ days and } \phi < 1 \text{ (stationary)}$$

The upper bound

- A very long half-life is a capital trap.
- The spread reverts too slowly to be worth it.
- Reject anything above about 30 days.

The hard floor

- A coefficient at or above one doesn't revert.
- There's no half-life - it's non-stationary.
- Reject it outright, whatever else it shows.

Four Pairs, Four Half-Lives

Screen the candidates

pair	AR(1) phi	kappa = $-\ln(\text{phi})$	half-life	verdict
A	0.85	0.163	4.3 days	ACCEPT
B	0.95	0.051	13.5 days	ACCEPT
C	0.99	0.010	69 days	REJECT (too slow)
D	1.005	-	none ($\text{phi} \geq 1$)	REJECT (no revert)

Pairs A and B revert fast enough to trade. Pair C reverts, but far too slowly. Pair D has a coefficient at or above one - it doesn't revert at all.

Applying the Filter

Accept two, reject two

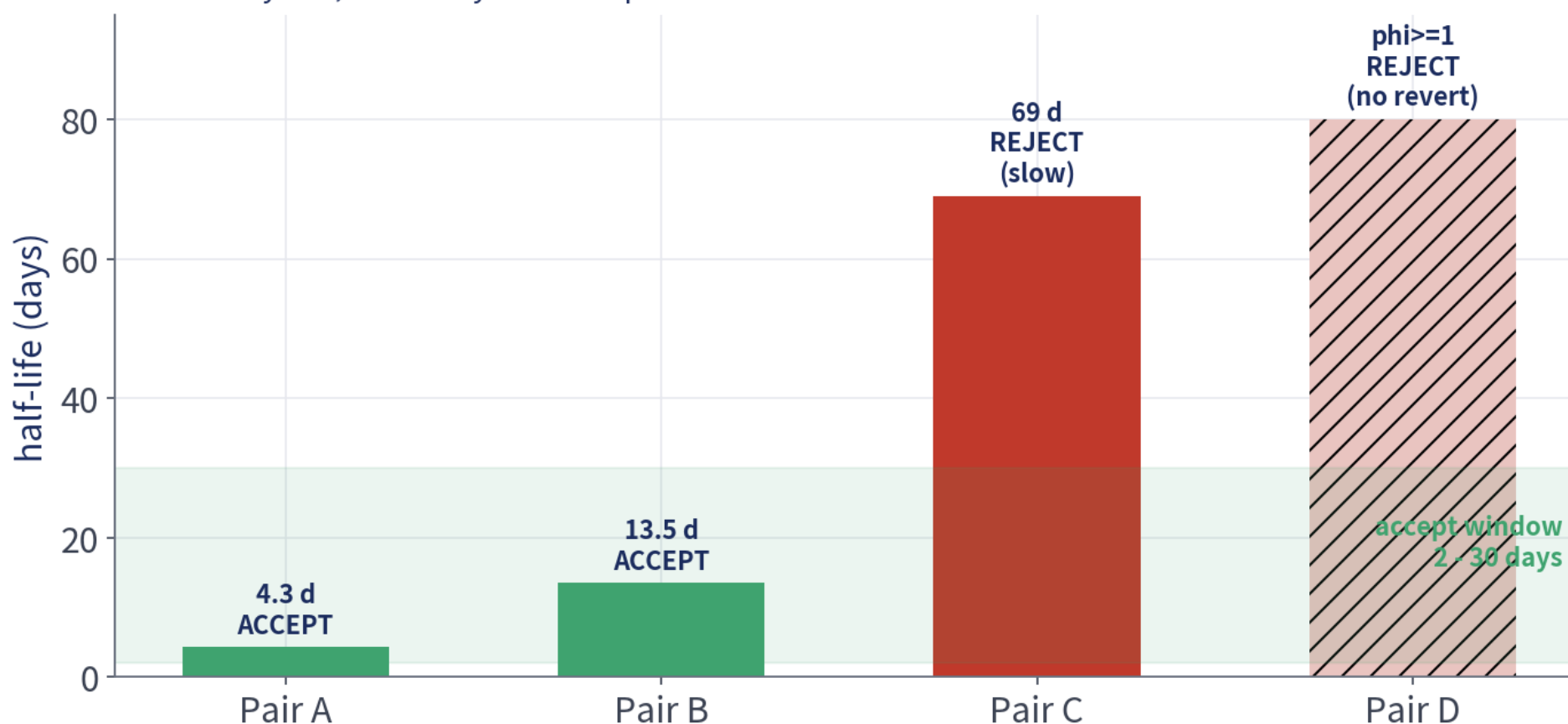
A ($\phi = 0.85$) : HL = 4.3 days $\Rightarrow$ **ACCEPT**
B ($\phi = 0.95$) : HL = 13.5 days $\Rightarrow$ **ACCEPT**
C ($\phi = 0.99$) : HL = 69 days $\Rightarrow$ **REJECT** (too slow)
D ($\phi = 1.005$) : $\phi \geq 1$ $\Rightarrow$ **REJECT** (not reverting)

The filter cleanly separates the portfolio: the two fast reverters go in, the capital trap and the non-stationary pair stay out.

Only Fast Reverters Pass

Pairs A and B fall inside the 2-to-30-day accept window and go into the portfolio. Pair C's 69-day half-life is a capital trap, and Pair D has a coefficient at or above one - no reversion at all. The half-life screens them in one glance

Case A: only fast, stationary reverters pass the half-life filter



Build the Portfolio Deliberately

The half-life is your first gate

Why screen at all

- Most candidate spreads aren't tradeable.
- Slow reverters lock up capital for thin returns.
- Non-stationary ones will just drift away.

What the filter buys you

- A portfolio of pairs that actually turn over.
- A single, comparable number per pair.
- A first line of defense against bad trades.

The Trade Economics

Why the half-life is the money number

Two pairs can offer the same profit per trade yet very different returns - because the faster one recycles capital more often. A round-trip takes a few half-lives, so a short half-life means more trades per year. We compute the annualized return for a fast and a slow reverter with the same 2% edge

The inputs

- Same 2% gross profit per round-trip.
- A round-trip runs about three half-lives.
- Pair A at 4.3 days, Pair B at 13.5.

What we'll do by hand

- Turn the half-life into trades per year.
- Multiply by the per-trade profit.
- Compare the two annualized returns.

Trades Per Year

Turnover comes from the half-life

$$\text{trades/yr} = \frac{250}{\text{cycle}} \approx \frac{250}{3 \times \text{half-life}} \quad (\text{faster reversion} \Rightarrow \text{more turns})$$

The cycle

- A trade reverts in about three half-lives.
- That's the capital-commitment time.
- Divide the trading year by it.

The consequence

- A short half-life means many trades a year.
- A long one means only a handful.
- Turnover is set by reversion speed.

Fast vs Slow, Annualized

The same edge, very different returns

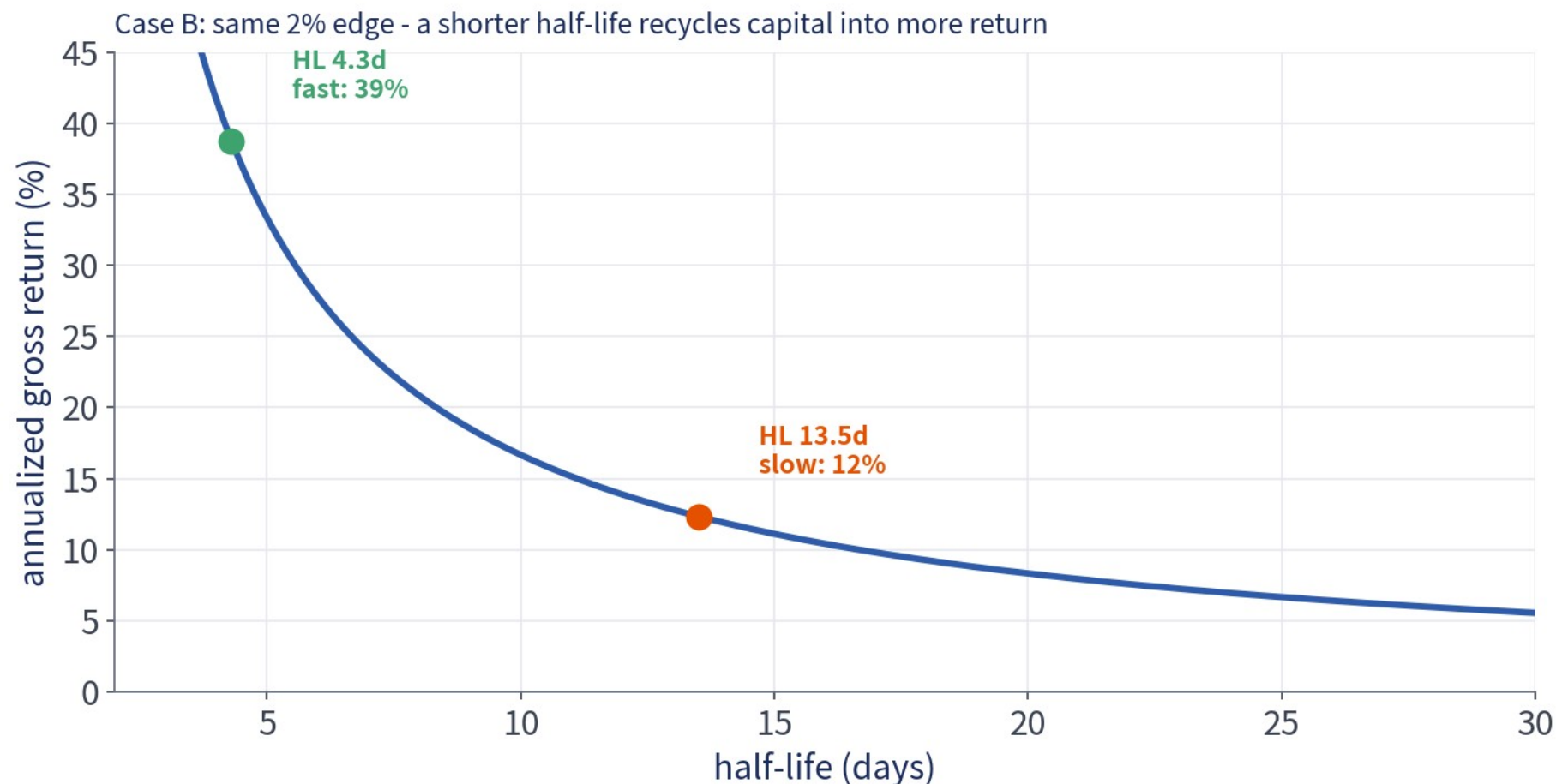
$$\begin{aligned} \text{fast (HL 4.3): } & \frac{250}{12.9} = 19.4 \text{ trades} \times 2\% = \mathbf{39\%} \\ \text{slow (HL 13.5): } & \frac{250}{40.5} = 6.2 \text{ trades} \times 2\% = \mathbf{12\%} \end{aligned}$$

Both capture 2% per trade, but the fast reverter turns capital 19 times a year for 39%, versus 6 times for 12% - roughly three times the return from the same edge.

Shorter Half-Life, More Return

With a fixed 2% edge, the annualized return falls off as the half-life lengthens - because capital turns over less often. The fast reverter at 4.3 days earns about 39%; the slow one at 13.5 days only 12%.

Reversion speed, not the per-trade edge, dominates



Turnover Beats Edge

And the risk that undoes it all

The economics

- Return is edge times turnover.
- The half-life sets the turnover.
- A faster reverter compounds more often.

The risk

- A regime break stops the reversion entirely.
- The spread drifts and the trade bleeds.
- Re-estimate constantly; a broken pair has no half-life.

The Pairs Journey, Now Five Stages

Volatility bands sized the trade (W5), cointegration anchored the spread (W7), a Kalman filter kept the hedge live (W9), a regime gate decided when to trade (W10), and now an OU model gives the spread a principled mean-reversion structure (W11). Stage V is principle over heuristics.

The Pairs Journey, Now Five Stages

Stage V replaces threshold heuristics with a principled OU model of the spread.



Stage V: principle over heuristics

The half-life screens pairs, the stationary band sets thresholds, and OU ties bands, holding time, and risk into one model.

One Number Runs the Portfolio

The half-life, everywhere

Stage V hangs on the half-life. It screens which pairs are worth trading, it sets how long each trade takes, and through turnover it drives the annual return. The OU model turns a spread's own statistics into a full trade specification - and reminds you to keep re-estimating because a broken spread has no half-life at all

Screening (Case A)

- Half-life filters the candidate pairs.
- Fast and stationary get in.
- Slow and non-reverting stay out.

Economics (Case B)

- Half-life sets trades per year.
- Turnover times edge is the return.
- Faster reversion, more compounding.

Reproduce the Screen and the Economics

Direct Claude Code to check your arithmetic

GOAL Verify the half-life screen and the trade economics, then build a small pair book.

[INPUT] Prompt · Claude Code

Write and run `part3_practice.py` that:

CASE A - screen:

- * for $\phi = 0.85, 0.95, 0.99, 1.005$, print the half-life and ACCEPT/REJECT (window 2-30 days, $\phi < 1$);

CASE B - economics:

- * with a 2% edge and cycle = 3 half-lives, print trades/year and annualized return for HL 4.3 and 13.5 days.

Then fit OU to several real spreads and rank them by half-life into a small pair book.

Match every printed number to the slides.

Check against the slides

- HL: 4.3, 13.5, 69 days.
- A, B accept; C, D reject.
- fast: 19 trades -> 39%.
- slow: 6 trades -> 12%.

Deliverable: `part3_practice.py` whose printed numbers match the screen and the economics.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 Why screen pairs by half-life?

**To keep only fast,
stationary reverters**

Q2 Why does a shorter half-life earn more?

**It recycles capital
into more trades a
year**

Q3 What kills an OU trade?

**A regime break - the
spread stops
reverting**

All three come from the half-life filter and the turnover economics.

Three Pitfalls to Avoid

Where Stage V slips

Common mistakes

- "Trade any spread that reverts eventually."
- "A bigger per-trade edge is all that matters."
- "Fit the half-life once and trust it forever."

The reality

- A slow reverter is a capital trap - screen it out.
- Turnover matters as much as edge for the return.
- Spreads break; re-estimate and stay humble.

Two Calculations to Remember

Part 3 applications in one screen

1

The half-life screens the portfolio

Of four candidates, A (4.3 days) and B (13.5 days) passed the 2-to-30-day filter; C reverted too slowly at 69 days, and D had a coefficient at or above one, so it doesn't revert at all. One number, four verdicts.

2

Turnover turns edge into return

With the same 2% per-trade edge, the fast reverter turned capital 19 times a year for 39%, while the slow one managed 6 turns for 12% - roughly three times the return, purely from a shorter half-life.

3

Stage V completes the pairs thread

Volatility bands, a cointegration anchor, a Kalman hedge, a regime gate, and now an OU model. The half-life screens pairs, sizes trades, and drives returns - principle in place of heuristics.

Next — Week 12: from a single stochastic process to systems, as the course turns to multivariate and machine-learning methods.